Chapter 5

SYNMAGMATIC DEFORMATION AND EXTENSIVE CRUSTAL CONTAMINATION OF THE PLATINIFEROUS UPPER CRITICAL ZONE AT THE MARGIN OF THE BUSHVELD COMPLEX: A FIELD STUDY IN DRAAIKRAAL, MPUMALANGA, SOUTH AFRICA

The Rustenburg Layered Series of the Bushveld Complex in the Mpumalanga, Limpopo, and Gauteng provinces of South Africa is amongst the world's best examples of a large, layered mafic-ultramafic intrusion and has been the focus of decades of research, yet the tempo and the fundamental nature of its formation remain open questions. Field research at key localities in the Bushveld Complex has provided invaluable insights into the emplacement and evolution of these layered intrusions, yet the intrusion is vast; many areas with good exposure remain poorly studied. Draaikraal is an agricultural town located in the eastern limb of the Bushveld Complex in the Mpumalanga province that sits at the platinum-rich intersection of the Critical and Main Zones with the intrusion's margin, presenting a valuable opportunity to study the formation geometry and timing of large mafic intrusions in general, with particular emphasis on its famous Platinum Group Element (PGE) mineralization.

Draaikraal is nested within a conspicuously recessed valley capped by ironstone laterites and hosts several unusual topographic and structural features, including the Kliprivier Synform and the Draaikraal Lineament (named here). The Kliprivier Synform represents a package of modestly rotated, locally pegmatitic plutonic rocks with conspicuously absent modal layering, and country rocks that include fine-grained, melanocratic Marginal Zone sills intercalated with migmatized quartzite and calc-silicate hornfels. The country rocks appear to be displaced over 1 km into the plutonic RLS near the mapped intersection of the Critical and Main Zones. The Draaikraal Lineament is a linear feature of recessive topography stretching from

the country rocks beyond the intrusion's margin to the middle Main Zone. It overlaps with the Kliprivier Synform and separates typical, resistant Main Zone gabbronorites to the north from the recessive Main Zone gabbronorites to the south in the agricultural valley hosting Draaikraal. Using structural measurements and correlation of mottled anorthosite marker beds of the lower Main Zone across the Draaikraal Lineament, I argue that the mottled anorthosite marker beds are not displaced across the Draaikraal Lineament as previous studies have suggested. I show that the Kliprivier Synform was deformed synmagmatically by Bushveld magmatism, and argue that it represents a monocline that formed due to laccolithic deformation of the country rock roof by the intrusion of middle Critical Zone sills below the level of the UG2 chromitite and overlying Merensky Reef and Main Zones, which appear to be undisturbed. The orientation of the Draaikraal Lineament is very similar to the surface-projection of the axial plane of the Kliprivier Synform, which likely continues at depth. The relative recessive topography of the Draaikraal Lineament and the Main Zone to the south is argued to be the result of higher degrees of contamination by hydrothermal fluids than the more resistant Main Zone gabbronorites of the surrounding area. The rocks south of the Draaikraal Lineament are envisaged to be fluxed with upwardly migrating fluids derived from devolatilization of underlying, relatively shallow sediments at the floor of the magma chamber whereas the resistant Main Zone to the north was underlain by thick Critical Zone rocks and thus was less contaminated by crustal fluids.

Author contributions. This chapter is being prepared for publication with co-author J. Roberts (University of Pretoria). **B.J.T.:** conceptualization, investigation (field mapping, sampling, and photography at Draaikraal), formal analysis, visualization, writing – original draft. **J.R.:** conceptualization, resources (hosting at the University of Pretoria and field logistics), investigation (field work), writing – review and editing. Generative AI was used as described in the front matter.

5.1 Introduction

The Bushveld Complex of South Africa in the Mpumalanga, Limpopo, and Gauteng provinces hosts the largest layered mafic intrusion in the world (the Rustenburg Layered Series, or RLS), which in turn bears the world's largest source of platinum group elements (Lee, 1996) as well as important reserves of chromium, vanadium, titanium, and iron (Cawthorn, 2015). It is exceptionally well-preserved, making it an ideal natural laboratory for studying the geochemistry and petrology of layered mafic intrusions. Even so, the order, tempo, and nature of the formation of the Bushveld Complex remain hotly debated (W. Kruger and Latypov, 2020; Mungall et al., 2016; Scoates et al., 2021). Classically, the RLS was thought to have been emplaced as a cohesive, long-lived magma chamber (F. J. Kruger, 1994; Cawthorn, 2015), but recently many have hypothesized that its formation was more complex, with incremental assembly through sill-like intrusions (Mitchell et al., 2019; Yao et al., 2021; Scoon and Mitchell, 2023; Bourdeau et al., 2023). The RLS is divided into the Marginal, Lower, Critical, Main, and Upper zones (von Gruenewaldt et al., 1985), which are often hypothesized to have formed in distinct events. Thus, the intersections of these stratigraphically defined zones at the intrusion's margin are particularly informative locations for detailed field work that seeks to unravel the emplacement history of the RLS. This field study focuses on several unusual but understudied structural and geomorphic features occurring at the economically significant intersection of the Main and Critical Zones with the intrusion's margin in the well-exposed Eastern Limb near the village of Draaikraal, Mpumalanga, South Africa: the Kliprivier Synform, the Draaikraal Lineament, and the unexpected recessive topography of the Main Zone in the agricultural valley surrounding the village of Draaikraal (van der Merwe, 2007; Molyneux, 2008).

5.2 Geologic Setting of the Draaikraal Area

A broad survey of the literature on the Bushveld Complex is presented in Appendix B. Here, the geologic setting in the vicinity of Draaikraal is described.

Previous work near Kliprivier and Draaikraal

Kliprivier was the focus of several early works focusing on the margin of the RLS to better understand the relationship between the RLS and the sedimentary country rock. van Biljon (1949) studied the hornfelsized carbonates

of the area and their relationship to the margin, observing an increase in metamorphic grade to wollastonite skarns near the margin of the RLS and reporting that the transgressive RLS mafic rocks are continuous along strike with the metasedimentary rocks, a finding that I also report here (see §B.5, "The Transformation Hypothesis" for elaboration). He used this and similar pieces of field and geochemical evidence to develop his provocative but unpopular Transformation Hypothesis: claiming that the RLS (and potentially other igneous rocks) were formed from in-situ magmatic-metasomatic reaction of sediment (see critique and discussion within the comparatively well-cited work of Cousins, 1959).

Kliprivier and the surrounding De Berg region were studied in detail by Frick (1967), who distinguished different facies of marginal sills, documented variable assimilation of country rock by marginal sills and the presence of Critical Zone and two chromitite seams in the area, and characterized contact metamorphism within the sedimentary rocks. Frick (1967) emphasized the discordance of the contact between the sediments and the RLS and reported that chromite seams are disaggregated at the contact and not present in the Marginal Zone. Button (1976) reports that the extent of the metamorphic thermal aureole decreases southward in the area as the RLS package thins.

In their influential paper geochemically categorizing the Marginal Zone sills, Harmer and Sharpe (1985) included a broad map of the Marginal Zone along the Bushveld's Eastern limb in addition to a detailed map of the marginal rocks at the boundary of the Main and Critical Zones at the Kliprivier Synform, described more in the "Structural Observations" section. These maps show that the field area at Draaikraal includes two distinct classes of Marginal Zone sills. The large map shows "B2" gabbroic sills to the north abutting the Critical Zone and "B3" gabbroic sills to the south abutting the Main Zone (although later works argue that the upper Critical Zone continues south within and beyond the field area: van der Merwe, 2007; Latypov et al., 2022). Furthermore, fine gabbros of the "B2" and "B3" lineages are also subdivided into "B2N" and "B3N". Because both "B2N" and "B3N" sills are magnetic and visually identical in outcrop they are both described here with the lithologic code "Rz", for fine-grained Marginal Zone. Their coarse-grained "B2" gabbros may (in part) correspond to the Disturbed Gabbronorite (Dcg) lithology mapped in this work.

van der Merwe (2007) mapped and described the geology on Kliprivier and several nearby farms along the eastern lobe of the RLS to assess the area's PGE resource potential. He first described the Kliprivier "Syncline", further detailed in the structural observations, unit descriptions, and discussion in this work. Within, he references several private studies which were contracted by mining companies that are not publicly available. Van der Merwe reported a core log from the area that is helpful to interpret field data.

Latypov et al. (2022) did a systematic but coarse field study of seven separate traverses across ~60 kilometers of the lower Main Zone in order to correlate the mottled anorthosites of the lower Main Zone, arguing that these correlated units, supposedly laterally continuous across a 4 km change in depth of the RLS floor, imply the existence of a kilometer-scale melt column. They argue their correlation requires a voluminous ~10^5 km^3 magma chamber during the emplacement of the Main Zone.

Setera and VanTongeren (2018) conducted a systematic study of the Main Zone in the Draaikraal area, reporting major and trace elements of major cumulus phases. They focused on characterizing magma recharge at the level of the Pyroxenite Marker in the upper Main Zone, but they also used chromium concentrations in pyroxene to identify a less significant magma influx at the level above the mottled anorthosites. This geochemical cross-section was compared with a similar cross-section to the north performed in an earlier work by VanTongeren and Mathez (2013), which notably featured more primitive Main Zone rocks than those observed near Draaikraal. Setera and VanTongeren (2018) argue that this southerly evolving compositional gradient within the Main Zone arose from evolution of migrating melts away from the feeder of the Bushveld, interpreted to be at the Thabazimbi-Murchison Lineament.

Topographic 'Inversion' and the Draaikraal Lineament

The agricultural village of Draaikraal in the Mpumalanga Province of South Africa sits in a local sedimentary basin. Geologically, it is situated within the lower Main Zone of the RLS. The Main Zone is known to be highly resistant to erosion and dominates surface exposure of the RLS in the comparatively poorly exposed North and West Limbs of the RLS. This is thought to be because of the common observation of inversion of small pigeonite grains into larger, continuous OPX grains, serving as a sort of resistant "cement", contributing to

the effectiveness of Main Zone rocks as dimension stone commonly used as a building material (Cawthorn, 2015). Figure 5.1a shows that this considerable change in topography at Draaikraal from the northern mountains to the southern valley occurs despite the continuity of the Main Zone through the lineament.

5.3 Methods

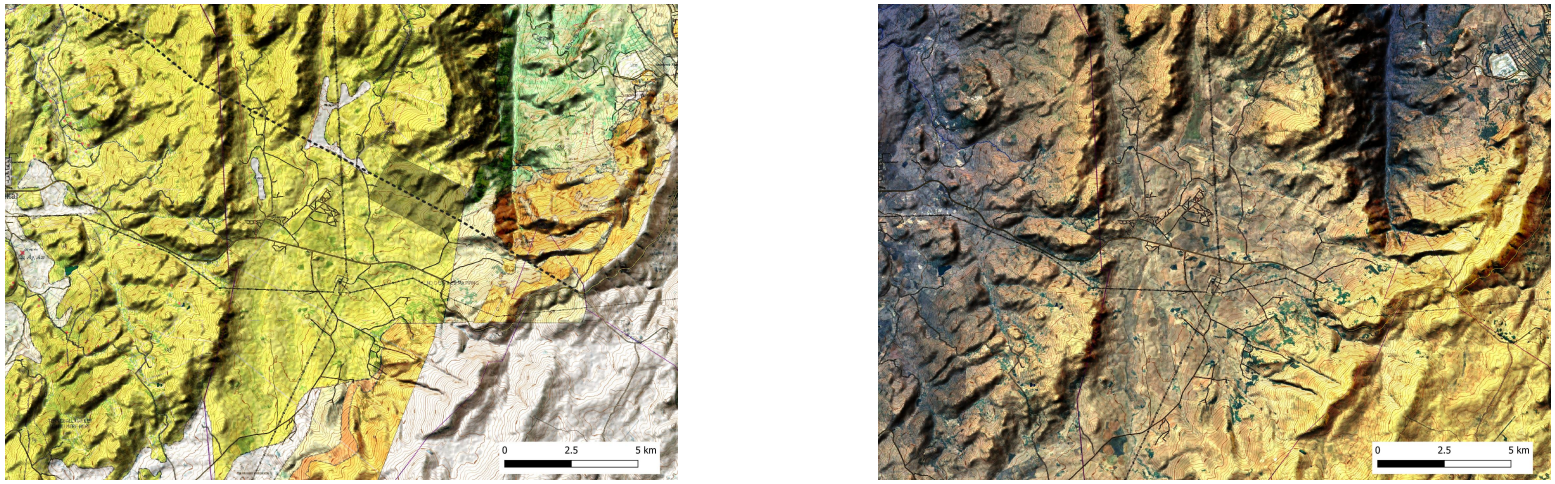
Field Methods

Field observations were made during an 11-day field campaign in July 2023. Field investigation and sampling strategies were adapted as the study proceeded, but initially were focused on correlating distinct rock units across the Draaikraal Lineament and taking structural measurements on each side to characterize it. Geolocated samples were taken across the field area for later geochemical work. The observation that plutonic igneous rocks to the north of the Kliprivier Synform were highly unusual and distinct from those to the south prompted a campaign to more thoroughly characterize its lithologic context and structure. Field observations, samples, and structural measurements were all geolocated and are recorded in Table A.1 (Appendix A). Almost every entry included an interpreted lithological unit to inform the final geologic map of the area (Figure 5.3). Many entries included photographs, which included mostly close-up photos of outcrops, but also some photos of the surrounding landscape to illustrate the area's geomorphology. Field photographs are cited throughout this chapter by number (e.g., photo 95); the photographs that are reproduced in the photo album of Appendix A (§A.2) are linked from their numbers, and the rest are in the supplementary photo album. **[TODO: Link to photo supplement here.]**

Mapping Criteria

The contact between the layered series (Critical and Main Zones) and the Marginal Zone is often gradational, requiring a somewhat arbitrary categorization. Since the Lower Main and Upper Critical Zones are non-magnetic and the finer-grained marginal zone rocks are magnetic, I chose field-detectable magnetism as the simple discriminant to distinguish marginal rocks from the massive Main and Critical Zones of the RLS.

When Layered Gabbronorites (Lg) are observed in the same outcrops as Mottled or Spotted Anorthosites, the locality is assigned to be Mottled or Spotted



(a) (b)

Figure 5.1: (a) The field area around Draaikraal with the geological mapping of Molyneux (2008). Due North is up on this map. The generally resistant, topographically high Steenkampsberg Quartzite and hornfels are mapped as orange units on the eastern margin. The Marginal Zone is mapped as the beige color between the quartzite and the RLS. The Main Zone is featured in yellow, the Critical Zone in the NE quadrant is green and exhibits strikingly recessive topography. A regional SW-NE trending fabric is prominent, which agrees with all structural measurements of rock cleavage taken in the lower Main Zone and upper Critical Zone aside from the rotated units associated with the Kliprivier Synform. The dark box outlines an area where SW-NE rock fabrics appear continuous across the Draaikraal Lineament (dashed line). The Main Zone around Draaikraal is unusually recessive relative to the Main Zone north of the Draaikraal Lineament, or to the west. The compilation of Molyneux (2008) did not consider more recent mapping that revealed the presence of the Upper Critical Zone to the south, terminating beyond the map area (van der Merwe, 2007). (b) The satellite view of the same area as (a), without geology and visual guides for clarity. The Draaikraal Lineament remains clearly visible. Elevations are colormapped such that relative lows are blue and highs are yellow. The Main Zone is generally at higher elevation than the Critical Zone (in the northeast quadrant) and the Upper Zone (to the west), while the quartzitic Steenkampsberg Mountains in the southeast quadrant are locally the highest feature.

Anorthosite.

Structural Measurements

All measurements of plutonic rocks reported here are orientations of near-planar rock cleavage (§5.4). Strike and dip measurements taken from the Kliprivier Synform were plotted on a stereonet and analyzed using Stereonet 11 (Allmendinger et al., 2012; Cardozo and Allmendinger, 2013). Poles to bedding planes of quartzites, hornfels, mafic sills, and plutonic rocks were calculated and plotted together in stereonet projection in Figure 5.7a. In Figure 5.7b, the fold axis is calculated using the poles to the bedding planes. The axial plane also was calculated, assuming it bisected the obtuse angle formed by the limbs. The obtuse fold angle bisector for the axial plane was selected because the rotation within the fold observed is quite gentle. An acute bisector for the axial plane would have been better supported if field evidence showed dramatic changes in dip along the fold, requiring a complete structural overturn that locally annihilated the original horizontality of the deformed rocks in addition to steeply dipping rocks intermediate to the two limbs.

Stratigraphic Column Analytical Methods

The field data of this study north of the Kliprivier Synform are projected onto a stratigraphic column following the assumption that all rocks in the field area north of the Kliprivier Synform follow Steno's Law of Original Horizontality and were only uniformly deformed by tilting due to isostatic flexure towards the center of the RLS after emplacement, so that all regional bedding and structures are currently dipping 15 degrees to the northwest relative to their original orientations. Geolocated field data are projected onto a line to the north of the Kliprivier Synform designed to be orthogonal to the regional strike (trending 301.79 degrees: W-NW) using the 'Snap Geometries to Layer' function in the 'Vector Geometry' Processing Toolbox in QGIS, with a tolerance of 1 degree set to "prefer closest point". Section heights then were calculated for these projected points along the reference line to an arbitrary point along this line, to get relative stratigraphic heights using the "Distance Matrix" function in the "Vector Analysis" Processing Toolbox. A correction was added to these lateral distances to account for elevation, calculated as the product of the elevation differential and the tangent of the regional dip, assumed to be 15°. The column heights are normalized to the height of the Merensky Reef. Because

the Merensky Reef was not observed in the field area, this was instead done by proxy: stratigraphic heights were calculated relative to the UG2 chromitite observed in the field area (§5.4) at a fixed height of -86 meters. This chromitite is mapped by van der Merwe (2007) as the UG2, who profiled drill cores and determined the Merensky Reef to be 86 meters above the UG2 in the field area (van der Merwe, 2007). Note that, for simplicity, it is assumed that there are no local changes in strike and dip, even though the dip of the Main Zone likely shallows and rotates to be roughly due east further from the margin of the intrusion. This approximation is likely to be acceptable locally for the upper Critical and lower Main zones, but becomes less accurate further from the margin. Therefore, the stratigraphic heights presented here for the Homogeneous Gabbronorite lithology far upsection are likely overestimated. The interpreted schematic stratigraphic column for the region is graphically presented in Figure 5.2.

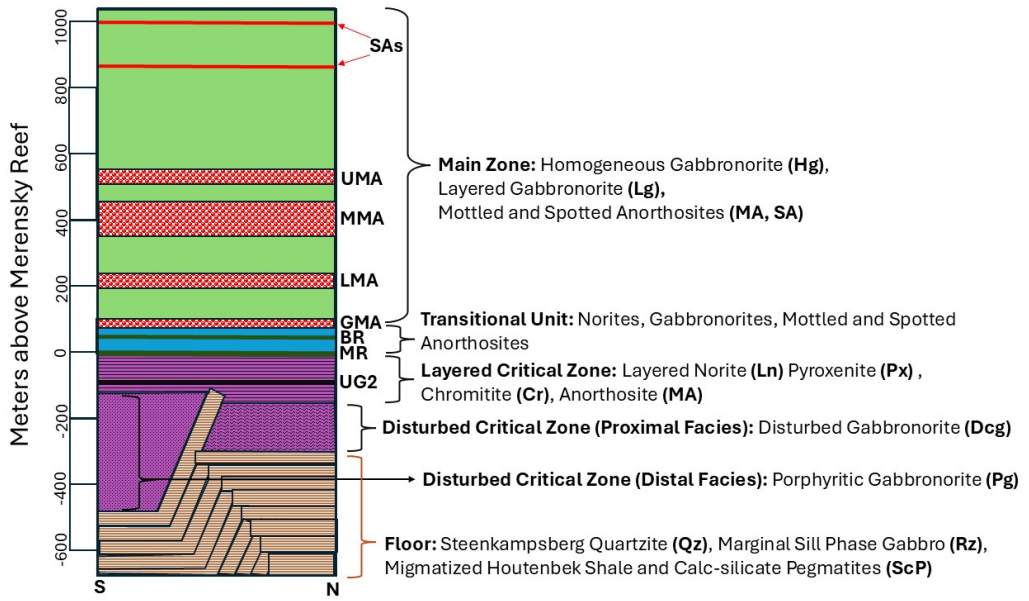


Figure 5.2: The schematic stratigraphic cross section of the RLS in Draaikraal. The Kliprivier Synform is represented by the slab of floor rocks protruding into (and creating) the Disturbed Critical Zone at the halfway point of the column. The Draaikraal Lineament would form from upward percolating fluids trapped and channeled by this protruding slab of floor rocks in a line directly above the slab in this view. The Kliprivier Synform is not interpreted to have deformed the RLS above the UG2.

5.4 Results: Field Observations

Observations are organized into three sections: lithologic descriptions, rock cleavage and modal layering, and structural observations; the first and last each have their own map.

Lithologic Descriptions

Lithologic descriptions are organized by the interpreted map units (see Figure 5.3) and their constituent lithologies. Three mapping units are described in turn, each with a wide variety of lithologies: the marginal rocks (the floor and walls of the intrusion, including sedimentary country rock and concordant pre- and syn-Bushveld magmatic sills); the plutonic rocks of the RLS; and secondary, post-Bushveld rocks that were largely ancillary to the field study.

Margin

Much of this study focused on the eastern margin of the eastern limb of the Rustenburg Layered Series (RLS) east of Draaikraal. The margin displays the most unambiguous bedding planes of the study area, generally dipping 25° to the northwest towards the center of the Bushveld Complex, except for the Kliprivier Synform, which is detailed below in the "Structural Observations" section. The marginal rocks within the field area are composed of gabbroic intrusive igneous sills concordantly intercalated with variably metamorphosed, melted, or otherwise devolatilized Transvaal sediments of the Steenkampsberg and Houtenbek formations. Occasionally, melting and/or reactive porous flow of melts through sedimentary layers remineralizes sedimentary layers with igneous mineralogy and texture (photos 35, 51, and 52). Concordant sills of marginal zone magmas and packages of sediment range in thickness from 10 to 100s of meters, and can be found mutually interbedded at least a kilometer from the margin of the RLS. Here, the marginal rocks are divided into three units: "Rz", the fine-grained Marginal Zone gabbroic sills/magmas; "Qz", relatively undisturbed Houtenbek quartzite; and "ScP", hornfelsed metapelites, marbles, and quartzites with related igneous/hydrothermal pegmatites.

Rz – Fine-grained gabbro

The Marginal Zone gabbroic sills are commonly fine grained (<1 mm), but locally are coarser with some grains up to 2 mm in the groundmass. Unlike

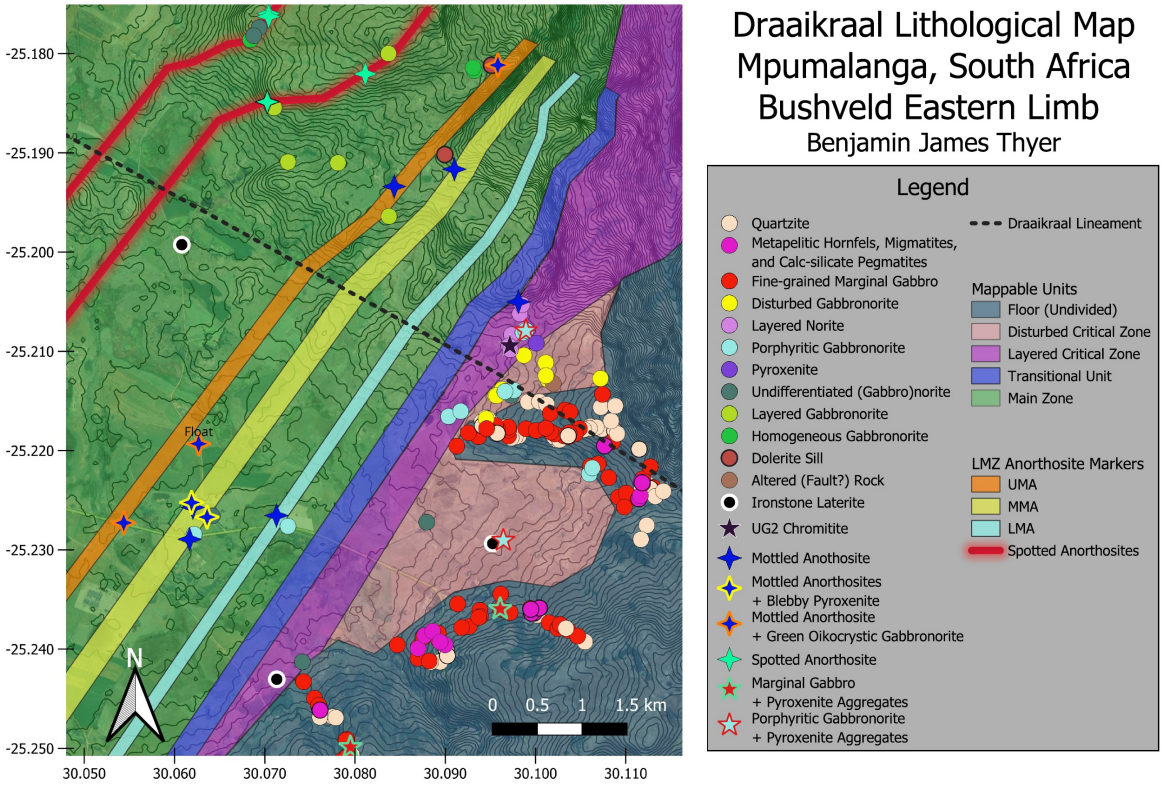


Figure 5.3: Lithological map of the Draaikraal field area, eastern limb of the Bushveld Complex. Field observations of outcrop are colored by lithology over the mappable units (floor, Disturbed Critical Zone, Layered Critical Zone, Transitional Unit, Main Zone). The lower Main Zone anorthosite marker beds (LMA, MMA, UMA, and the spotted anorthosites) are traced in color, and the Draaikraal Lineament is dashed. Field observation data are listed in Table A.1.

much of the plutonic RLS, the Marginal Zone is not visibly modally layered. Marginal Zone magmas within a meter of Steenkampsberg Quartzite often contain conspicuous evidence of silicic contamination: although they retain an igneous texture, the color significantly lightens and the rocks become much harder and less magnetic approaching the contacts with Steenkampsberg Quartzite. Aside from these immediate contacts, the Marginal Zone gabbro is always magnetic. The Marginal Zone contains more pyroxene (distinguishing clinopyroxene and orthopyroxene is difficult with such small crystals) than plagioclase with variable magnetite, sometimes up to 10%, and minor phlogopite. The heavy, fine fraction of mineral separates from one sample was briefly studied under optical microscope to identify trace phases, which include apatite, zircon, sulfides, and euhedral chromite.

The Marginal Zone magmas contain igneous xenoliths and are rich in paleo-vertical, across-strike features that likely record rising, buoyant fluid/magma. Perfect planar resistance banding (photo 95) is the most common structure in the fine grained Marginal Zone and is nearly ubiquitous within the unit. Though the resistance bands have seemingly random strikes, they most often are steeply dipping and may represent fluid conduits along brittle fractures within the Marginal Zone.

In previous works, isolated meta-sedimentary xenoliths (Buick et al., 2000) were emphasized, but in this study, meta-sedimentary rocks were never found as minor xenoliths within magnetic marginal sills, but rather as continuous, concordant strata (photos 42, 70, 71) or as locally abundant disaggregated broken strata cemented within a pink, evolved diatexite (photo 100), both designated as a distinct unit: ScP.

The magnetic sills of the Marginal Zone (Rz) did include enclaves of coarse-grained leucogabbronorite, mottled anorthosite or pyroxenite (photos 2,4,6,96). The anorthosite-rich enclaves are rounded, 6-10 cm along their longest dimension, and are imbricated to be preferentially lying flat paleohorizontally within the Marginal Zone, suggesting they were weathered/eroded by reaction and/or mechanical transport through Marginal Zone magma. Pyroxenitic enclaves are also often spherical (photos 4, 96) and may be related to the formation of the pyroxenite “Tennis Ball Marker” (TBM) of the Main and Critical Zones, described further in the discussion. These plagioclase-rich igneous xenoliths contain large (<3 cm) optically continuous oikocrystic augite that enclose

blocky plagioclase (<5 mm).

More commonly observed within the magnetic Marginal Zone sills east of Draaikraal are paleovertical structures that suggest buoyant upwelling of fluids and melts sourced from the hornfelsized sedimentary rocks. These structures included sheared, viscously deformed lenses of thin pegmatites, including anorthosite (photo 6), magnetite gabbros (photo 60, sheared gabbro ~15 cm; Figure 5.8), and evolved quartz-biotite pegmatites (photo 67: 1×3 cm). In addition to homogeneous pegmatites, the Marginal Zone gabbros host paleovertical invasions of anastomosing anorthosite derived from anatexis of underlying metasedimentary rocks (photos 6, 97). These leucosome intrusions occasionally carry blocks of meta-sedimentary paleosome or brecciate and enclose the Marginal Zone gabbronoritic host rock itself (photo 98).

Near the contact with the migmatite, differential weathering sometimes reveals proud, thin, pinched ovate paleovertical structures (5×1 cm scale) on weathered faces of Marginal Zone norite that are otherwise invisible on a fresh rock face [photo 40].

Fine-grained Marginal Zone gabbros are mapped up to 120 meters below the Merensky Reef stratigraphically.

Qz – Houtenbek Quartzite

The Houtenbek quartzite dips concordantly with marginal zone sills and migmatized sedimentary rocks towards the center of the Bushveld, here dipping about 25° to the northwest. Even within meters of a contact with Marginal Zone sills (Rz), the quartzite often retains a clastic sedimentary texture and possibly could be considered a proper clast-dominant quartzose sandstone with well-sorted and well-rounded coarse quartz grains ~1 mm in diameter with no other mineral visible. While there is little structure visible internally because of the purity of the quartzite, the bedding planes occasionally reveal conspicuous asymmetric ripple structures [photos 2, 68] <1 cm in height. One in-place ripple structure suggests a paleoflow direction towards the south-southwest. Some quartzites are hornfelsized: recrystallized into an interlocking crystalline texture with large, irregular, angular, coarse grains <2 cm across. Many quartzites with preserved clastic textures near the margin of the RLS are stained bright red within rocks' unweathered interior. This staining is hypothesized to be derived from mobilized Fe-rich fluids percolating along grain boundaries, probably

caused by heat and/or fluid circulation related to the emplacement of the RLS. Steenkampsberg Quartzite is mapped up to 140 meters below the Merensky Reef stratigraphically.

ScP – Houtenbek Hornfels, Migmatites, Pegmatites

In addition to quartzite, the Houtenbek formation contains carbonate and pelitic horizons (Button, 1976). At the margin east of Draaikraal, there is a 60 m thick package of melted and metamorphosed sediment containing stratiform pegmatites with variable mineralogy. Hornfelsized sediments come in two main varieties near Draaikraal: pink, diatexite-supported breccias [photo 100] or dark, high-grade schists [photo 29] with thin but conspicuous navy, gray, white, tan, and pink bands of relatively mafic and felsic minerals that have been recrystallized and no longer bear clastic textures except for thin seams of clast-dominated quartzite in a low-fraction glassy biotite-rich matrix [photo 35, large light seam is ~4 cm thick]. This unit is rarely magnetic and most often is visible as continuous, concordant strata [photo 42]. Even when highly brecciated, this unit can be followed along the strike of the quartzites and Marginal Zone, implying that the sediments were not incorporated into Marginal Magmas, but that they rather were melted and devolatilized in-situ due to heating from the emplacement of adjacent marginal sills. Fluids and melts produced in this process are visible as a wide variety of stratiform pegmatites and skarns with undulating erosive contacts within comparatively planar meta-sediments [photo 30: feldspar/wollastonite(?) + amphibole skarn; photo 69: magnesite, quartz pegmatite with minor siderite] and as pockets following fluid flow pipes and void spaces that are accommodated by the host strata through ductile deformation [photo 37, 38: Orthopyroxene and quartz pegmatite]. Hornfels in the field area are mapped up to 190 meters below the Merensky Reef stratigraphically.

The Rustenburg Layered Series (RLS)

The Rustenburg Layered Series is the world's largest individual igneous intrusion (Cawthorn, 2015; Latypov et al., 2022) and is subdivided into the Lower Zone, Critical Zone, Main Zone, and Upper Zone (§B.2). In this study, RLS rocks of the lower Main Zone, Transitional Zone, and upper Critical Zone are mapped. The rocks of the Marginal Zone (Rz) are not layered, nor considered here

to be a part of the RLS on a textural basis. Marginal zone rocks weather distinctly from the plutonic RLS phase with abundant, sharp, ~3 cm-scale potholes and generally are more textured and rough [photo 36] on a weathered surface than RLS rocks, which weather spheroidally and proudly [photos 8, 11: "porphyritic" gabbronorite]. Xenolith blocks of Marginal Zone norites bearing their distinctive resistance banding are found in both the Critical and the Main Zones [Critical Zone: photo 76; Main Zone: 91], though each xenolith sighting within the Main Zone is proximal to a mottled or spotted anorthosite layer. These xenoliths are variably annealed, recrystallized, and even ductily deformed [Critical Zone: photo 77; Main Zone: 84].

The RLS is famous for its distinctive and numerous laterally continuous stratiform marker horizons, sometimes stretching on a 100-km scale (Cawthorn, 2015), which include the Critical Zone chromitite cycles, Merensky and Bastard pegmatoid reefs, three mottled anorthosite packages in the lower Main Zone, the Pyroxenite Marker of the upper Main Zone, and the magnetitite cycles of the Upper Zone. In this study, I observe and describe the three lower Main Zone magnetic mottled anorthosites [photos 49, 53, 86, 93, 105, 106, 108, 111], the magnetic chromitite UG2 [photos 65, 66], and a non-magnetic upper Critical Zone mottled anorthosite that is probably a part of the Merensky Reef footwall [photo 81], and several packages of xenolith-bearing modally layered gabbronorites and spotted anorthosites within the Main Zone (photos 91 and 111). These laterally continuous reefs are mapped and described separately from the bulk cumulate rocks of the RLS.

Cz: Critical Zone

The Critical Zone was previously mapped to pinch out due east of Draaikraal (shown in the compilation of Molyneux, 2008); however, many now recognize that a thin layer of the upper Critical Zone persists much further south towards Tontledoos (van der Merwe, 2007). In this study, diagnostic Critical Zone lithologies were only observed north of the Kliprivier Synform, where typical finely modally layered norites and anorthosites are well-exposed (photos 64 and 80), as well as the UG2 chromitite. The Critical Zone above the Kliprivier Synform dips ~15° to the northwest similarly to the majority of the Main and Marginal Zone rocks in the field area, but gradually rotates to a southwestern dip as it approaches the contact with the rotated block of metasediments and

gabbronorite sills of the Kliprivier Synform. It is essential to note that some rotated Critical Zone rocks are structurally beneath the large, rotated package of intruded metasediments (Rz + Qz + ScP) in the area of the Kliprivier Synform, meaning that this package of country rock is not only serving as the floor for RLS rocks to the west, but is also (at least locally) a roof for Critical Zone rocks below the level of the UG2.

Dcg: Disturbed Gabbronorite This lithology is restricted to the immediate area of the Kliprivier Synform. Disturbed Gabbronorites are coarse-grained, even pegmatoidal (photo 62), with feldspars and pyroxenes up to 3 cm long. This unit was named in response to van der Merwe (2007), who said that the "structural change [of the Kliprivier Synform] is reflected in the Bushveld rocks, which become highly disturbed and fragmented". Disturbed Gabbronorites are most recognizable for their abundance of blobby autoliths of anorthosite (photo 61) and pyroxenite (photos 70 and 72). While it exhibits abundant examples of modal separation, they are never remotely planar, unlike the majority of the RLS.

This lithology is most often non-magnetic, but a few specimens were magnetic. Besides the mottled anorthosites of the Main Zone and the anorthosite-hosted chromitite leader of the UG2, no other rocks of the RLS within the field area are magnetic. Many but not all Disturbed Gabbronorites are rotated within the Kliprivier Synform. Generally, the Disturbed Gabbronorites are CPX-poor and sulfide-rich with abundant, euhedral <1 cm pyroxenes set in an anorthosite matrix that effectively isolates the large, euhedral pyroxenes from one another [photo 62]. This unit generally becomes more feldspathic as one moves away from the intruded metasediments of the Kliprivier Synform into the Critical Zone. This lithologic unit contains recrystallized and annealed xenolith blocks [photos 74, 75, and 77] of fine-grained Marginal Zone gabbros (Rz) as well as coarse-grained pockets containing very large (< 5cm) oikocrystic pyroxene [photo 73] and large clusters of pyroxene often rimmed by thick anorthosite halos [photo 72]. Disturbed Gabbronorite is mapped between 120-145 meters below the Merensky Reef stratigraphically, around the level of the UG1 in core logs (van der Merwe, 2007). van der Merwe (2007) and Latypov et al. (2022) report isolated patches of massive chromite here, but I did not see any.

UG2: Upper Group 2 Chromitite Near the top of the upper Critical Zone in the extreme northern portion of Kliprivier farm, there is a small but high-quality exposure of chromitite (>30 cm thick). The bulk of the exposure is a melanocratic, non-magnetic, anorthositic chromite, which is modally ~50% <1mm euhedral chromite set within a fine matrix of <1 mm plagioclase (photo 66). Structurally above this massive layer are some thin (<1 cm thick) chromitite leaders hosted within bright white mottled anorthosite (photo 65). Both layers contain amoeboid oikocrystic pyroxene-olivine mottles (<2cm). The mafic silicate mottles and anorthosite matrix seem to have crystallized within the interstices of a preexisting framework of a continuous chromite crystal slurry [photos 65, 66], (see Voordouw et al., 2009; Mukherjee et al., 2017, for similar interpretations). The massive melanocratic layer includes pure (relatively chromite-free) white anorthosite blocks (<4×15 cm) that are lying flat and dipping west towards the center of the Bushveld Complex. These pure anorthosite blocks contain thin chromitite stringers that together have been ductilely deformed, and a thin (~1 mm) white seam of pure, apparently undeformed anorthosite forms a plane bounding the top of some of these blocks, also dipping to the west.

The anastomosing chromite leaders within the overlying mottled anorthosite are sometimes highly magnetic and are a much deeper, vitreous black: appearing under hand lens to be >80% modally chromite/magnetite [photo 65].

This chromitite package is interpreted as UG2 for a few reasons. First, although the anastomosing chromite leader resembles the UG1 stringers, the leader is structurally above the massive chromitite, whereas the UG1 stringers are recognized below the massive chromitite (Voordouw et al., 2009; Mukherjee et al., 2017). The UG2 is known to have several small "leaders" a small distance above the main seam (Maier and S.-J. Barnes, 2024). Second, the magnetic chromitite within the leader bears explaining. One heterogeneous magnetic chromitite at the top of the UG2 with discrete magnetic and non-magnetic bands was described by Frankel (1942), 20-30 km to the north of Kliprivier at Tweefontein. Third, a rock at this location was also mapped as the UG2 by Harmer and Sharpe (1985, Fig. 4) and van der Merwe (2007, Fig. 3), who also identified the UG2 and closely overlying mottled anorthosites in drill cores of the area using PGE grades. van der Merwe (2007) also reports that the UG2 at Kliprivier is unusually enriched in iron, which may explain the magnetism

of the leader. The UG2 is 86 meters below the Merensky Reef in the direct vicinity of the Kliprivier Synform (Borehole 3: van der Merwe, 2007).

Ln: Layered Norite: Spectacular outcrops of modally layered norites within the upper Critical Zone are observed within the field area at Kliprivier around and above the stratigraphic level of the UG2 chromitite (photos 64 and 80), at stratigraphic heights from 100 to 50 meters below the Merensky Reef. Modal layering is pronounced, with pyroxenite-rich layers being separated by near-pure centimetric anorthosite bands. This layering is undulose and irregular.

Px: Pyroxenite This refers to rocks predominantly composed of pyroxenes. Pyroxenites are rare in the field area. There was one coarse-grained sulfide-bearing feldspathic pyroxenite horizon in the Critical Zone just below the UG2 (photo 78) as well as a pyroxenite layer just below a mottled anorthosite layer in the UMA in the lower Main Zone (photos 53–54).

Tz: Transitional Zone

There is no perfect division for the Main and Critical Zones. Moving upsection from the Critical Zone into the Main Zone, Cr_2O_3 concentration within OPX decreases, strontium isotopes become more radiogenic, and cumulus clinopyroxene becomes important— but these changes occur at different stratigraphic heights within the transition zone (Cawthorn, 2015) between the Merensky Reef and Giant Mottled Anorthosite (or above). Within the literature, the bottom of the Main Zone is considered to be above the Giant Mottled Anorthosite at the top of the Bastard unit, and the Merensky reef is taken to be the top of the Upper Critical Zone (Cawthorn, 2015; Mitchell and Manthree, 2002; van der Merwe, 2007). The Bastard and Merensky pyroxenites and Giant Mottled Anorthosite were never observed in the field area, but a ~50 cm thick mottled anorthosite [photo 81] was observed ~50 meters upsection from the UG2 with only finely layered norite [photo 80] and porphyritic gabbronorite outcropping between the UG2 and the mottled anorthosite. Since van der Merwe (2007) reports in core logs that the Merensky Reef is 86 meters above the UG2 on Kliprivier and Latypov et al. (2022) observe unusual pyroxenites just above this mottled anorthosite, this anorthosite layer is interpreted to be a part of

the footwall to the Merensky Reef, and the pyroxenites described in Latypov et al. (2022) are interpreted to be the Merensky Pyroxenite.

Pg: Porphyritic Gabbronorite The "Porphyritic Norite" is a well-recognized marker of the lower Main Zone, with large, rounded pyroxene phenocrysts set in a white, feldspar-rich matrix, used often in mining to locate the Merensky Reef (SACS, 1980; von Gruenewaldt et al., 1985). I observe a general textural similarity in lower Main Zone Gabbronorites with the unlayered upper Critical Zone gabbronorites below the Merensky Reef (photos 10, 79). These gabbronorites are distinct from the dark and homogeneous gabbronorites that compose the majority of the Main Zone (Hg). The porphyritic gabbronorite contains large ($\sim<$ 1cm), equant pyroxenes set in a leuconoritic matrix that is predominantly plagioclase with small spots of interstitial orthopyroxene. I refer here to them collectively as the "porphyritic gabbronorite" because I see cumulus clinopyroxene present throughout, becoming more modally abundant upsection until about half of the cumulus pyroxenes are CPX near the Lower Mottled Anorthosite (LMA). Generally, cumulus CPX is present in the Main Zone and absent from the Critical Zone (Cawthorn, 2015). However, some cumulus CPX has been reported at the top of the Critical Zone (Cameron, 1982; SACS, 1980).

The rocks of this unit have well developed cleavage [photo 63]; however, they are not visibly layered. I located porphyritic gabbronorite south of the Kliprivier Synform at the beginning of the second field day and assumed it was Main Zone given the cumulus CPX I saw, but now I believe that the porphyritic gabbronorite occurs in the upper Critical Zone through the Transition Zone and into the lower Main Zone in the vicinity of Draaikraal. The Porphyritic Gabbronorite in this field area hosts spherical pyroxenite tennisballs (photos 10, 79), in the same locations as mapped by Harmer and Sharpe (1985) and Mitchell and Scoon (2023). Porphyritic Gabbronorite is mapped in the Critical and Main Zones 480 meters below the Merensky Reef to 380 meters above, stratigraphically.

Mz: Main Zone

The Main Zone is the most volumetrically significant portion of the RLS: as thick as 5 km (Latypov et al., 2022). The Main Zone stratigraphy is (in

general) relatively featureless aside from three prominent mottled anorthosite packages in the lower Main Zone and the Pyroxenite Marker at the upper Main Zone, which was not studied in this work. Mottled anorthosites and associated strongly modally layered sequences are mapped and described here in order to study the effect of the Kliprivier Synform and Draaikraal Lineament on the stratigraphy, as these unique layers can be traced great distances through the Lower Main Zone (Latypov et al., 2022).

Hg: Homogeneous Gabbronorite: The massive Homogeneous Gabbronorite of the Main Zone varies slightly in color due to variations in the modal abundance of pyroxenes and plagioclase, though these variations are subtle and continuous on a scale of 10s of meters. There is no obvious fabric visible in outcrops or hand samples of these massive rocks, although I was shown a polished core of upper Main Zone gabbronorite from a dimension stone mine east of Roossenekal that obviously displayed a fabric where crystals preferentially were aligned with the gravitational plane. Stonemasons favor this rock for building, but are careful to allow only faces cut against this fabric to outwardly show for aesthetics and durability. Homogeneous Gabbronorite (Hg) begins about 450 meters above the Merensky Reef and continues upsection stratigraphically through the Pyroxenite Marker (SACS, 1980; von Gruenewaldt et al., 1985).

(L/M/U)MA: (Lower/Middle/Upper) Mottled Anorthosites Mottled Anorthosites are enigmatic igneous rocks composed of a nearly pure, white anorthosite matrix containing (often quite large: <5 cm) mossy green "mottles" of oikocrystic pyroxene +/- olivine that enclose abundant plagioclase. In the RLS they are found in large stratiform horizons interpreted to be continuous across the majority of the Main Zone in the Bushveld (>100 kilometers: Latypov et al., 2022). The lower Main Zone near Draaikraal contains 3 large mottled anorthosite packages. These mottled anorthosites are magnetic and notably contain rare large (< 2mm) grains of magnetite within the mottles. They are found in association with larger packages of distinct but non-magnetic anomalous lithologies. The associated unusual rocks include: 1) fine green gabbronorites with blebby, anhedral phenocrystic (<2 cm) clinopyroxene (lithologic code Fgog: Fine-grained oikocrystic gabbronorite) associated with the UMA; 2) feature-rich, coarse pyroxenites (lithologic code: Px, [photos 53, 54]) and

3) blobby pyroxenitic gabbronorites (lithologic code Bap: Blobby anorthosite-pyroxenite) that include deformed, irregular, decimeter-scale blobby pyroxene agglomerations within an anorthositic matrix that are interpreted here to be recrystallized blocks [photos 104, 108]. Sometimes these blobby anorthosite-pyroxenites exhibit modal layering of coarse (<2 cm) pyroxene oikocrysts [photo 105] within anorthosite. The mottled anorthosites and their associated pyroxenites and blobby gabbronorites have undulating, nonplanar contacts with one another [photo 49] and with the underlying common gabbronorites [photos 107, 103]. Sometimes these contacts may be drastically discordant from the local strike and dip. In photo 106, the mottled anorthosite - gabbronorite contact is dipping ~70° to the east towards the margin of the chamber: a much steeper dip and opposite strike than observed in the leucogabbronorites of the immediate vicinity. Occasional 3D exposure of the anorthosite-gabbronorite contact allowed me to measure the contact rotating 90+ degrees in this outcrop within meters (photo 107). It is unclear if these undulating contacts are evidence for discordance of the mottled anorthosites, since in other places the contact appears concordant [photo 110], and local apparent discordance could be the result of thermal erosion of the cumulate pile by overlying magmas (Roberts et al., 2007).

In this study, all three lower Main Zone anorthosites described by Latypov et al. (2022) and mapped by van der Merwe (2007) were located within the low-lying valley south of the Draaikraal Lineament at stratigraphic heights between 200 and 600 meters above the level of the Merensky Reef. The Upper and Middle "mottled" anorthosites in the mountains to the north reported by Latypov et al. (2022) were located in this study, but are visually quite distinct and occupy higher stratigraphic heights, argued here to be distinct units that stratigraphically overlie the thick, massive mottled anorthosites in the valley to the south (see the Discussion of this work). von Gruenewaldt et al. (1985) report multiple subordinate anorthositic layers near and at the boundary of the Main Zone subzone A and B much higher above the level of the Main Mottled Anorthosite. Those higher anorthosites are referred to here as spotted anorthosites (SA) to reflect their distinct texture.

SA: Spotted Anorthosites North of the Draaikraal Lineament where exposure is better, two packages of anorthosites were found in the lower Main Zone. These anorthosites are somewhat distinct from the common, unlayered

gabbronorites of the Main Zone, but anorthositic beds are significantly smaller (<1 m thick) and altogether less visually striking than the Mottled Anorthosites (MA) unit described above. The spotted anorthosites are non-magnetic and bear <1 cm spots of equant pyroxenes composing <20% of the rock by volume [photos 86, 87] that sometimes contain recrystallized pyroxenite-gabbronorite blocks [photo 111]. The Spotted Anorthosites are strongly associated with much thicker (<10 m) packages of planar, finely layered gabbronorites that are unusual within the Main Zone and are more reminiscent of the Critical Zone [photos 85, 112]. These modally layered sections also include xenoliths [photos 91, 84], which were only observed within the Main Zone in these modally layered packages. These Spotted Anorthosite packages are argued here to be distinct and stratigraphically above the remarkable and prominent Mottled Anorthosites described above. Two or three packages of Spotted Anorthosites (often including thick layers of Layered Gabbronorite) are interpreted here at stratigraphic heights 700-1050 meters above the Merensky Reef. More packages may exist, although they are unlikely to be exposed to the south of the Draaikraal Lineament.

Lg: Layered Gabbronorite Modal layering is generally less common in the Main Zone than the underlying Critical Zone, and in outcrop it is comparatively understated. Whereas the Ln (Layered Norite) unit of the Critical Zone contains vivid, centimeter-scale, near-monomineralic anorthositic bands, the modal layering in the Main Zone gabbronorites is millimeter-scale, represented by subtle differences in cumulus grain size and modal proportions, and the contacts between layers are planar (photos 85 and 112). The Layered Gabbronorites are associated with Mottled Anorthosites (MA) or Spotted Anorthosites (SA). Layered Gabbronorites are mapped here at stratigraphic heights between 400 and 1050 meters in association with Mottled and Spotted Anorthosites.

Post-Bushveld Rocks

In the valley surrounding Draaikraal, alluvium primarily is composed of well-rounded quartzite cobbles (~10 cm) and ironstone laterites (photo 59). The ironstone laterites are commonly found as both float in alluvium and as in-place outcrops. Bushveld mafic rocks are rare in alluvium, probably due to their relative susceptibility to chemical weathering on Earth's surface. Quartzite was

substantially dominant over RLS mafic rocks in alluvium even 8 km away from the nearest known quartzite source within the RLS, leading Dr. James Roberts and me originally to suspect that there were closer quartzite sources; however, we were unable to identify quartzite sources west of Draaikraal. It is likely instead that the creeks which source their headwaters from the Houtenbek and Steenkampsberg quartzite highlands to the east and flow past Draaikraal to the west transported and rounded these cobbles, which are much more stable on the surface than the RLS rocks, eventually coming to dominate the regional alluvium.

IS: Ironstone Laterites Ironstone laterites are common throughout the relatively low and flat valley surrounding Draaikraal. They are somewhat weakly consolidated and highly porous non-magnetic amalgamations of <1cm iron concretions composed of 1 mm thick deep black vitreous goethite rims surrounding pockets of fine-grained yellow limonite matrix. Interspersed between these concretions are variably rounded medium-coarse Steenkampsberg quartzite gravels. The goethite nodules and quartz gravels are cemented by fine grained, red and yellow hematite-limonite matrix with mm-scale angular, coarse, quartz sand. The existence and abundance of these ironstone laterites implies that the valley surrounding Draaikraal is a longstanding, persistent topographical feature that has existed on geological timescales. They were likely formed by in-place weathering of RLS rocks, explaining the relative dearth of RLS material within the valley alluvium.

Dl: Dolerite Sill: Post-Bushveld Dolerite sills are mapped throughout the field area (van der Merwe, 2007; Molyneux, 2008), and one such discordant sill was observed at the levels of the UMA and MMA north of the Draaikraal Lineament within the field area (photo 89). Despite being younger than the surrounding RLS, the sill is significantly more weathered, likely because of extensive blocky fracturing hypothesized to result from rapid contraction during cooling. The dolerite is moderately magnetic.

Rock Cleavage and Modal Layering

Within many of the phaneritic plutonic rocks, there was measurable, near-planar cleavage visible (e.g. photos 58 and 63). Plutonic rock cleavages without modal layering yielded measurements that were similar to the regional dip

of bedding in the floor metasedimentary rocks to the east, but in areas with prominent modal layering these cleavages had strikes rotated 90 degrees from the regional strike, interestingly dipping similarly to the exotic limb of the Kliprivier Synform. These 90 degree offset structures could be measured very near to one another, sometimes within the same outcrop. Photo 115 shows rock cleavage that is discordant with modal layering in a striking outcrop of the MMA.

It is important to note that one outcrop of mottled anorthosite clearly showed disagreeing cleavage and planar modal layering, with modal layers dipping 50° more eastwardly than the rock cleavage [photo 115, annotated in Figure 5.4].

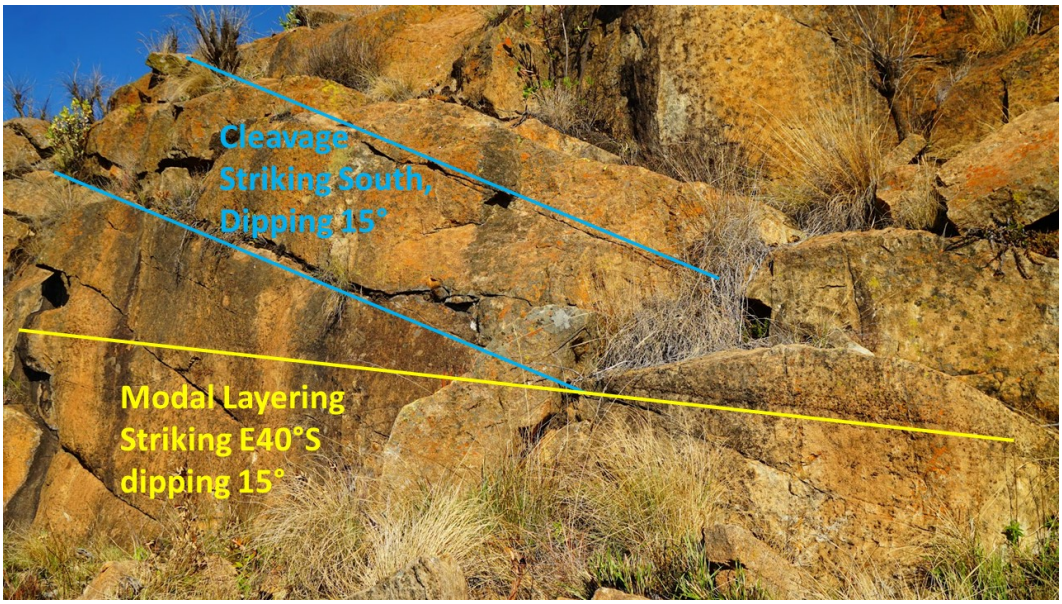


Figure 5.4: Cleavage and modal layering forming distinct planes in a Mottled Anorthosite outcrop (annotated photo 115).

Structural Observations

Structural measurements across the field area, together with the traced lower Main Zone anorthosite marker beds, are mapped in Figure 5.5.

Kliprivier Synform

The most interesting large-scale structure in the mapping area (Figure 5.6) is a rotated package of disturbed, pegmatoidal upper Critical Zone pyroxenites and gabbronorites [photos 61,62, 72–78], well-cleaved leuconorite layers [photo 63], quartzites [photos 28 and 124], migmatized/hornfelsed argillites and calc-silicate

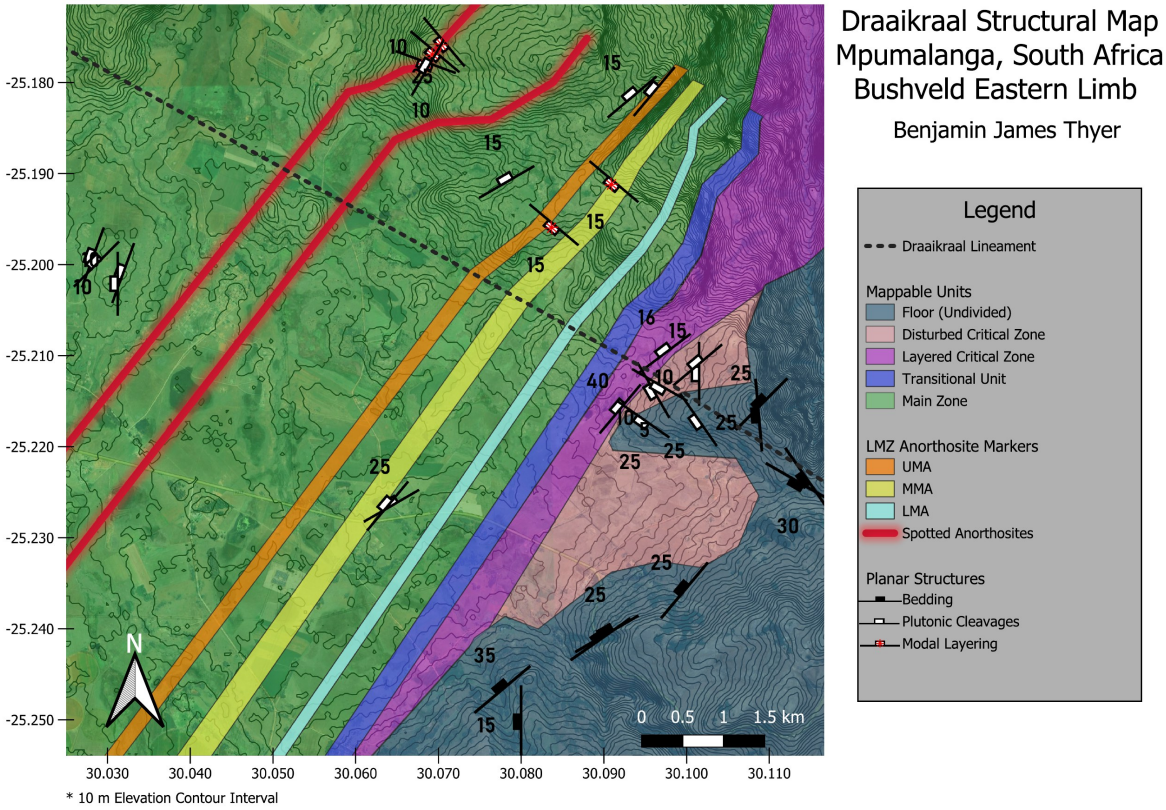


Figure 5.5: Structural map of the Draaikraal field area: strike and dip of bedding, plutonic cleavage, and modal layering, over the mappable units (floor, Disturbed Critical Zone, Layered Critical Zone, Transitional Unit, Main Zone), with the lower Main Zone anorthosite marker beds (LMA, MMA, UMA, and the spotted anorthosites) and the Draaikraal Lineament (dashed). 10 m elevation contour interval. Measurements are listed in Table A.1.

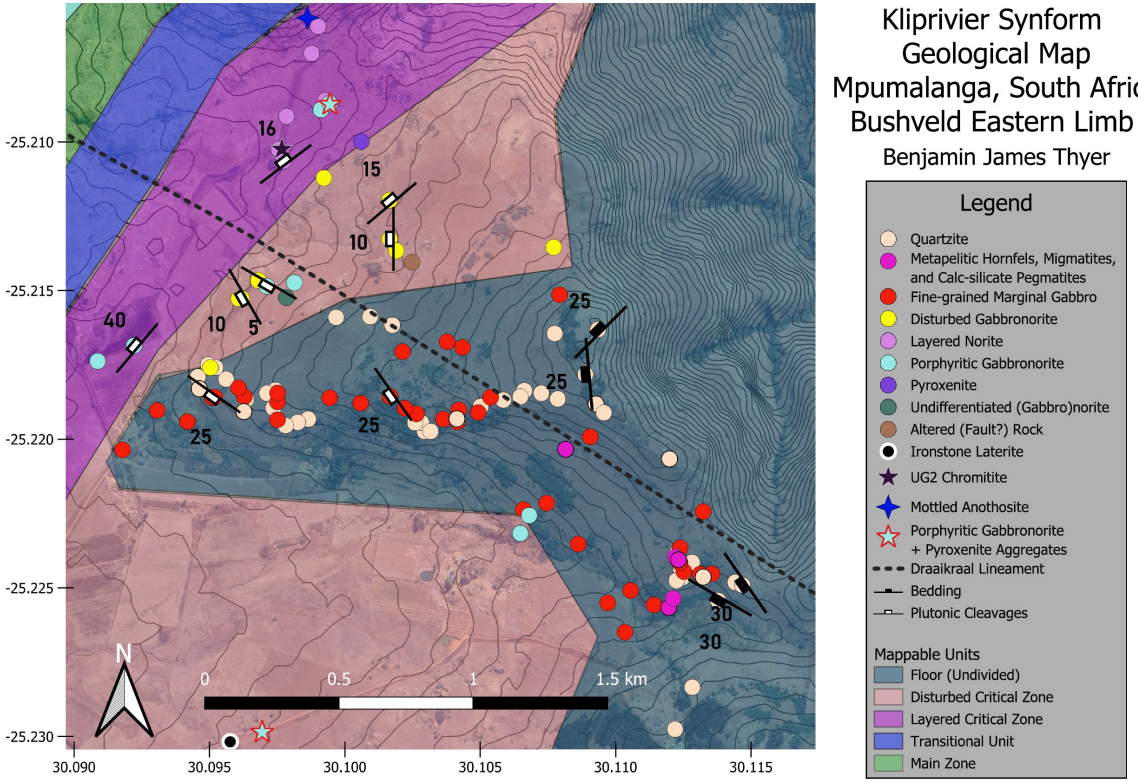


Figure 5.6: Geological map of the Kliprivier Synform. Field observations are colored by lithology over the mappable units (floor, Disturbed Critical Zone, Layered Critical Zone, Transitional Unit, Main Zone), with bedding and plutonic-cleavage measurements and the Draaikraal Lineament (dashed).

layers [photos 29, 30, and 125], and marginal sills [photo 57] in the northern portion of Kliprivier Farm. The structure is remarkable because it apparently displaces a marginal package of metasediments and fine-grained sills over 1 km into the plutonic RLS. This structure is described accurately but qualitatively by van der Merwe (2007), who investigated the area to assess its platinum resource potential. van der Merwe (2007) refers to this feature the "Kliprivier Syncline", though it is not clear if he is naming the structure or it was named elsewhere. A displaced quartzite layer within the structure was also observed during the field mapping of the Main Zone mottled anorthosites by Latypov et al. (2022) and is described in their supplementary information as a "Large Quartzite Xenolith?".

The structure is referred to here instead as the "Kliprivier Synform" to recognize that it is not yet well understood. For instance, no previous work to date has correlated marginal magmas and country rocks from the two limbs of the putative syncline to demonstrate that they are the same and that the structure is indeed a fold and not another structure, such as a laccolithically deformed roof remnant of an RLS subchamber or a package of floor rocks entrained in a rising magma. The map presented here offers additional details that elucidate the nature of this structure. Below, a series of observations of the Kliprivier Synform is presented:

1. To the north of the structure, there is a continuous, 90° gradual change in strike moving from finely-banded noritic Critical Zone, which strikes southwest and dips northwest like the majority of the mapped lithologies east of Draaikraal, to a southeastern strike dipping southwest. This gradual change in structure is accompanied by lithologic changes in the Critical Zone: Near and at the deformed areas, grain-size increases (photo 61) and stratiform modal layering ceases and is replaced by a heterogeneous rock with abundant fine-grained Marginal Zone xenoliths, anorthosite autoliths (photo 62) and irregular clumps of pyroxene surrounded by anorthosite (photos 72–77). I have mapped these lithologies as "Disturbed Gabbronorite" (Lithologic code: Dcg).

2. To the south, there is little exposure and no structural data could be gathered to pinpoint the reversal of the structure back to the common northwestern dip. However, outcrops constrain the contact between the

Marginal Zone magmas and the porphyritic gabbronorite of the upper Critical Zone to occur within 10 meters. The "disturbed" lithologies of the Marginal/Critical zones present on the north side of the synform are not present on the south side. van der Merwe (2007) interprets faulting with downthrows to the south at this contact; however, van der Merwe (2007) does not map any vertical displacement of marker beds such as the UG2 chromitite at this fault.

3. Stratigraphically higher plutonic rocks are undeformed, suggesting that they were intruded and crystallized after the deformation forming the Kliprivier Synform. Interestingly, the western termination of the synform as mapped here occurs at or slightly below the level of the Merensky Reef, where a significant magma recharge event corresponding to the emplacement of the Main Zone is thought to have occurred (Seabrook et al., 2005; Cameron, 1982).

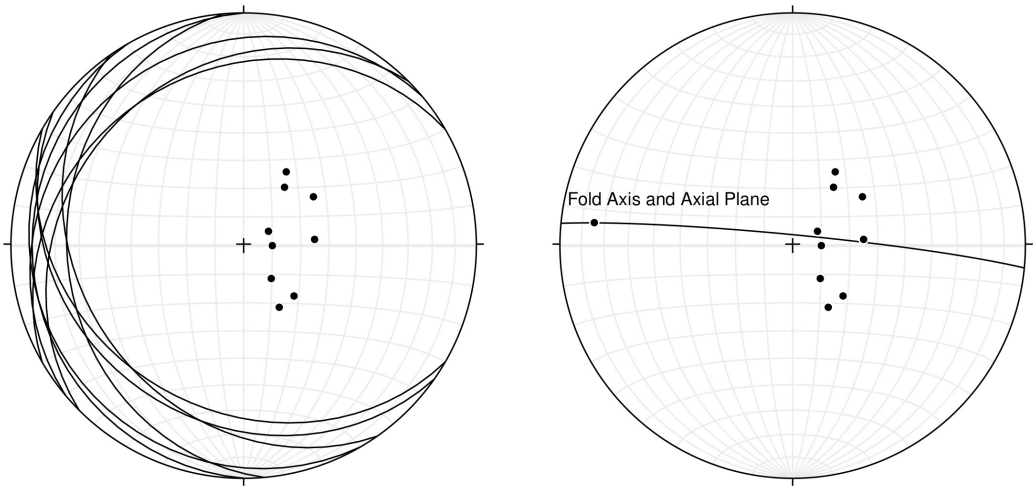
4. Displaced plutonic rocks to the north are stratigraphically below the quartzites and Marginal Zone sills, meaning that the section of sills and sediments serve as a roof to the RLS to the north.

5. Within this study, there is one strike and dip measurement immediately to the west of the structure, that is striking SW like the majority of the field area but is dipping more steeply than other rocks in the region. Latypov et al. (2022) also report an unusually steeply dipping plutonic rock along strike.

6. To the east, the structure is attached to the floor of the RLS. Unfortunately, that area was not mapped in detail and the nature of the rotation from the synform's exotic dip to the common dip of the marginal series was not observed.

The calculated fold axis (Figure 5.7b) trends roughly W5°N, plunging 15°, similarly to the Draaikraal Lineament. The calculated obtuse bisector axial plane strikes W5°N and dips 85°.

The axial plane is similar in orientation to the WNW-ESE trending Draaikraal Lineament that bounds the topographically high and low Main Zone rocks.



(a) (b)

Figure 5.7: Structural analysis of the Kliprivier Synform in stereographic projection (Stereonet 11; Allmendinger et al., 2012; Cardozo and Allmendinger, 2013). (a) Bedding planes and poles to planes of rocks within the Kliprivier Synform. (b) Calculated fold axis and axial plane for the Kliprivier Synform.

5.5 Discussion

Origin of the Kliprivier Synform and Draaikraal Lineament

van der Merwe (2007) described a "synformal structure" on Kliprivier farm and referred to it as the "Kliprivier Syncline", but I identified only a single, exotically-dipping limb within the structure with a gradual southerly change in strike, therefore I interpret it here as a monocline.

van der Merwe (2007) brings attention to the PGE grade of such structures along the eastern margin of the RLS, which often are rich in sulfides and contain unusually high-grade Pt ores aside from the traditional targets: the UG2 chromitite layer and the Merensky Reef (§B.5, "The Origin of the Merensky Reef and PGE Mineralization"). van der Merwe (2007) believes that the Kliprivier structure and these other sulfide traps predate Bushveld magmatism; however, I identified multiple coarse-grained plutonic rocks of the upper Critical Zone that were rotated within the structure. No Main Zone or Transition Zone rocks exposed on the surface to the west of the Kliprivier are deformed within the Kliprivier Synform, suggesting that deformation preceded the crystallization of the Main Zone. This relative ordering requires that the structure is synmagmatic. This interpretation is also supported by evidence for viscous deformation of gabbroic pegmatoids on the synform (Figure 5.8).

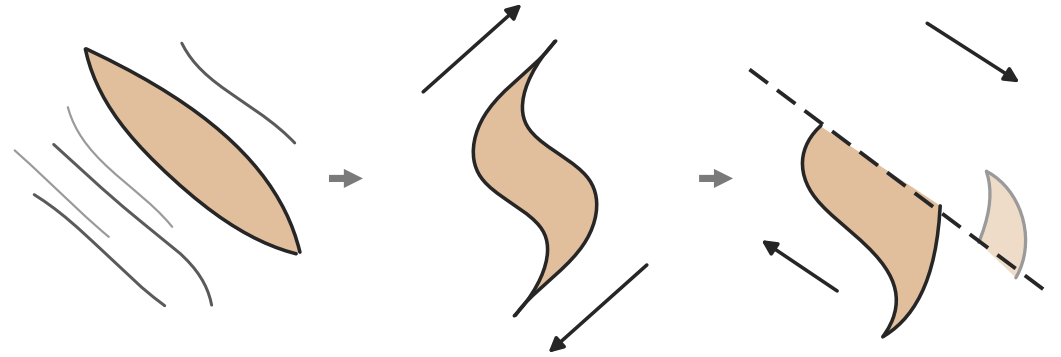
I interpret this unit to be formed by synmagmatic deformation of nascent layered upper Critical Zone mushes during the formation of the Kliprivier Synform, effectively breaking up and redistributing the nascent layering into many blobs.

Structural analysis of the synform is consistent with gentle folding along a fold axis plunging shallowly to the west with a subvertical axial plane trending roughly E(SE)-W(NW) bisecting the limbs' obtuse intersection angle (Figure 5.7b). This deformation is subtle and would not reverse the original ordering of the layered series and the marginal series. A syncline alone cannot explain how the Marginal Zone of the deformed, SW dipping limb came to be structurally higher than some of the deformed plutonic rocks. It is, however, consistent with the idea that much of the stratigraphically lower Critical Zone was intruded as stacked sills from the north and terminated here, laccolithically deforming the overlying country rocks (§B.3). Within this paradigm, the intruded Transvaal sediments "displaced" deeper within the RLS at the Kliprivier Synform are actually roof rocks to the Critical Zone below the level of the UG2. Given the intense melting and devolatilization of the floor rocks in the area, it would not be surprising to find that the thin strand of country rocks dividing the Critical Zone here beneath the level of the UG2 would have been completely reacted to igneous minerals and textures by reactive porous flow of magmas and hydrothermal fluids, as has been argued for some Lower Zone and Marginal Zone rocks (Yudovskaya et al., 2021). Indeed, the ghosts of such transformed strands may be expected at many locations throughout the complex and may have separated sills of distinct composition or served as traps for percolating melts or hydrothermal fluids. They may therefore be expected especially at geochemical discontinuities within the stratigraphic column, such as the Merensky Reef, Giant Mottled Anorthosite, Pyroxenite Marker, Upper Mottled Anorthosite, etc.

Sharpe and Chadwick (1982) also reported that Critical Zone rocks but not Main Zone rocks were deformed in the area, but attributed similar small-scale folding on the margin to isostatic flexure. Isostatic flexure has affected the modern expression of the Bushveld Complex immensely, but a paleomagnetic study shows that isostatic flexure only commenced within the Bushveld after cooling beneath the Curie temperature of magnetite: 575°C (Hattingh, 1986). This means that flexure had to occur after termination of Bushveld magmatism



(a)



(b)

Figure 5.8: (a) Sheared magnetite pegmatoid inclusion within non-magnetic marginal rocks, interpreted as reacted sediments, on the western side of the Kliprivier Synform (photo 60). The pegmatoid is about 15 cm long. (b) Interpreted deformation history: (1) mafic pegmatitic melt accumulates in lenses in migmatized sediment; (2) synmagmatic viscous deformation produces a sigmoid geometry; (3) post-magmatic brittle failure along an annealed fault truncates the lens and displaces its tip (lightly outlined). Black arrows show the sense of shear at each stage. The second stage of deformation may have occurred during isostatic flexure during cooling. The anneal of the fault suggests that it occurred at warm subsolidus conditions.

and cannot explain synmagmatic deformation.

If the southwesterly dipping limb of the Kliprivier Synform is rotated ~20° about a level rotational axis trending ~45° NE to "undo" the local effects of postmagmatic isostatic flexure that affected the RLS and its margin, the exotic limb associated with the Kliprivier Synform would be dipping ~20° due south. This observation is consistent with the hypothesis that an intrusion originating from the Thabazimbi-Murchison Lineament to the north terminated in this area, producing a southerly dipping monocline in its roof rocks.

The axial plane of the Kliprivier Synform is similar in orientation to the Draaikraal Lineament (§5.4). Considering that the Kliprivier Synform also rests upon this lineament, it is likely that the formation of the Kliprivier Synform is related to the Draaikraal Lineament.

Origin of Spheroidal Pyroxenite Aggregates (Tennis balls)

Spheroidal aggregates (SAs) of pyroxenite up to ~5 cm are known to occur as inclusions in the (gabbro)norites near the margin of the RLS, and larger pyroxenite "boulders" within the Merensky Reef are well-documented in the western limb of the RLS (Lee and Sharpe, 1979; Smith et al., 2023; Khoza, 2021). The spherical agglomerates (also called "tennis balls" due to their similar size) are richer in pyroxenes, higher in An# and pyroxene Mg#, and richer in quartz and sulfides than their (gabbro)noritic hosts (Lee and Sharpe, 1979; Khoza, 2021). Originally, dense groupings of these spherical aggregates were thought to represent laterally continuous marker beds, but they cannot be traced along strike of RLS layering for long distances relative to the named reefs or mottled anorthosites– instead they are most commonly found <200m stratigraphically above the margin with the floor rocks regardless of the stratigraphic position within the layered series (Lee and Sharpe, 1979). This author located isolated pyroxenite tennisballs in porphyritic leuconorite in two places above and below the Kliprivier Synform [photos 8, 79], on regions previously mapped to contain tennisballs (Lee and Sharpe, 1979). More interestingly, this author located spheroidal aggregates of pyroxenite in the fine-grained Marginal Zone [photos 2, 4, 96]. Only very recently have authors argued the marginal zone hosted spheroidal aggregates at all. Mitchell and Scoon (2023) used geochemical evidence to claim the coarsely crystalline plutonic gabbronorites hosting the Tennis Ball Marker (TBM), famously known on the Middelkraal farm, ought

to be considered as Marginal Zone and not as Main Zone as previously mapped (Lee and Sharpe, 1979; Molyneux, 2008; Khoza, 2021). Before this work, no fine-grained marginal zone rocks had been reported to contain pyroxenite tennis balls or spheroidal aggregates, although those observed here in Draaikraal are much more lumpy and irregular than the *spheroidal* aggregates (SA) found in the Middelkraal TBM.

In photo 96, the largest pyroxenite SA (~4 cm) resembles a diapir: It has a long (~15 cm) and thin (~<1 - 2 mm) pyroxenite tail developed behind it, suggesting that it was migrating through a comparatively rigid, stationary crystal-rich mush. The formation of a diapir suggests that the SA was moving relative to the host rock and that the host rock was more rigid than the pyroxenite. This requires explanation, as the spheroidal aggregates contain more primitive pyroxenes and plagioclase than the host matrix and are equilibrated at higher temperatures, meaning that the pyroxenitic SAs would be more rigid than the host rocks if one assumes the two are in thermal equilibrium.

There are three main hypotheses for formation of the spheroidal pyroxenite aggregates in the RLS. The model of Lee and Sharpe (1979) posits that pyroxenite aggregates are formed when a second, immiscible fluid called the "bridging" liquid intrudes a magma body with suspended pyroxene crystals. The bridging liquid preferentially "wets" the suspended pyroxene crystals, and entrains them as it coagulates with other smaller, individual bubbles of the immiscible fluid entraining the suspended pyroxenes of the host magma.

The second model of Khoza (2021) posits that a magma recharge event introduces magma into a pyroxenite cumulus pile, breaking the pile into pieces that are then subsequently entrained into the secondary magma. This magma does not equilibrate with the pyroxenite; instead, it evenly melts the rough edges of the pyroxenite aggregates, fashioning them into spheres.

The third model posits that a pyroxenitic liquid or mush intrudes a cooler crystal mush and is disaggregated into pyroxenite aggregates, analogously to the formation of mafic enclaves in felsic plutons from the disaggregation of intruded basalts (Mitchell and Scoon, 2023; Smith et al., 2023). This is the model most compatible with the observation of diapirism of spherical aggregates in the fine-grained marginal sill phase, since it allows for the SAs to be formed by disaggregation of a comparatively hot ultramafic magma intruded into comparatively cool (and therefore rigid) mafic sills.

The diapiric movement of the pyroxenite spheroid in photo 96 serves as good evidence against the second model of Khoza (2021), as Khoza's model requires the pyroxenite spheroids to be more rigid than the host magma, so that they remain consolidated against the flowing host magma. If the spheroidal aggregate was more rigid than the host magma, then it would not have left a pyroxenitic tail behind it. Finally, if magma recharge introduced into a pyroxenite cumulate pile forms SAs, then one would expect them to be a ubiquitous feature throughout the Bushveld Complex regardless of the distance from the margin, as the conventional and most accepted model for the assembly of the Bushveld Complex evokes many instances of magma recharge atop a preexisting cumulate pile (Holness et al., 2017; Cawthorn, 2015).

The model of Lee and Sharpe (1979) is interesting because secondary fluid input both can explain the relative softness of the primitive SA migrating through more evolved host rocks, and because it explains the abundant quartz (>10%: Lee and Sharpe, 1979) in the pyroxenite SA, which would not be expected for an ultramafic magma with high Mg# and An#. Photos 97 and 98 exhibit the migration of a distinct secondary comagmatic silica-rich fluid through the marginal zone magmas. However, Mitchell and Scoon (2023) point out that multiple distinct SA-rich horizons of the TBM exist but those that they report were over 100 meters from the alleged source of the fluids: the Dullstroom lavas at floor of the magma chamber. Their model simply requires injection of hot pyroxenite lavas into cooler mafic mushes. This author, however, suggests that the secondary bridging liquids of Lee and Sharpe's model could have percolated through homogeneous (gabbro)norites until ultimately meeting a coarse, pyroxenitic horizon susceptible to differential wetting.

The model of Lee and Sharpe (1979) offers a satisfactory explanation for the SA's ties to the intrusion's margin. The secondary fluids that circulated throughout the Bushveld chamber were predominantly sourced from devolatilized and variably melted country rocks (Schiffries and Rye, 1990; Buick et al., 2000, see also §B.4, "Stable Isotopes of the RLS"), so Bushveld magmas near the margin would be the most likely to encounter poorly mixed, newly-introduced immiscible (or poorly miscible) fluids than magmas further from the margin. The model of Mitchell and Scoon (2023) also offers a plausible explanation for the SAs' ties to the intrusion's margin, as the margin would cool faster to create a larger temperature (and viscosity) contrast between new pulses of

pyroxenitic magmas and their host rocks.

Correlation of the Mottled Anorthosites in the Lower Main Zone

Although there are three main named mottled anorthosite packages within the lower Main Zone, there are also many other minor anorthosites throughout the Main Zone (SACS, 1980; von Gruenewaldt et al., 1985). There is agreement between van der Merwe (2007), Setera and VanTongeren (2018), and Latypov et al. (2022) about the location of the UMA on the Roossenekal-Lydenburg road. I allow this agreement to calibrate the naming of the area's mottled anorthosites. Additionally, van der Merwe (2007) and Latypov et al. (2022) also mapped the MMA and LMA along the Roossenekal-Lydenburg road at the same location as I have.

van der Merwe (2007) maps the surface exposure of the UMA, MMA, and LMA throughout the southern sector of the eastern limb of the RLS, and later worked with Latypov et al. (2022), who undertook 6 stratigraphic traverses throughout the southeast RLS to prove that these 3 marker beds are laterally extensive throughout the entire Main Zone and do not abut the country rocks, arguing that these continuous mottled anorthosite packages were emplaced at the floor of an entirely molten 5 km thick magma chamber. Three of these traverses overlapped with the field area of this work. Within the field area, the UMA and MMA mapped by van der Merwe (2007) and Latypov et al. (2022) are discordant; they do not follow the regional dip to the northwest. I did locate anorthosites at the coordinates provided in the SI by Latypov et al. (2022); however, I conclude that the UMA and MMA that they have mapped in their traverse I, north of the Draaikraal Lineament, are not the same anorthosites as their mapped UMA and MMA south of the Draaikraal Lineament on the Roossenekal-Lydenburg road in their traverse III. I further argue that what they have mapped as UMA and MMA north of the Draaikraal Lineament are actually spotted anorthosites and some lack mottles entirely.

Two claims should be distinguished here. The first, which I hold with confidence, is that the anorthosites correlated with the UMA and MMA north of the lineament by van der Merwe (2007) and Latypov et al. (2022) are spotted anorthosites: they are thin, non-magnetic, and lack the coarse oikocrystic mottles and associated pyroxenites that define the mottled anorthosites of the lower Main Zone (Table 5.1). The second, which is tentative, is which of the

northern mottled-anorthosite packages corresponds to the UMA and which to the MMA.

Though this particular contested correlation by Latypov et al. (2022) is of minor significance to the overall argument put forward in their work, it is essential evidence to understand the origin and significance of the Draaikraal Lineament as the surface expression of a synmagmatic monocline formed by laccolithic deformation during the intrusion of much of the Critical Zone (see the end of this subsection).

The outcrops behind both correlations are compared in Table 5.1; their photographs are in Appendix A.

The UMA outcrop on the Roossenekal-Lydenburg road (photos 53, 54) contains a distinct package of a striking magnetic mottled anorthosite, ~0.5m thick underlain by a ~0.5m coarse-grained feldspathic pyroxenite with some reacted anorthositic blocks and pure anorthosite bands. This pyroxenite is in turn underlain by a unique rock interpreted to be a fine-grained gabbronorite containing blebby anhedral bottle-green clinopyroxene phenocrysts. Fresh faces of this rock are a uniquely vivid green color. I was able to locate the vivid green fine grained gabbronorite with blebby anhedral pyroxene phenocrysts (Fgog) above the Draaikraal Lineament near a cliff with a view of the Booysendal PGE mine to the east (Fgog not pictured, photos 88 and 90). I correlate this outcrop with the UMA south of the Draaikraal Lineament because it preserves the same sequence of fine-grained green gabbronorite, coarse feldspathic pyroxenite and magnetic mottled anorthosite. It sits ~170 m lower in the stratigraphic column (398 m versus 564 m above the Merensky Reef; Table 5.1). The match is not perfect, but the column assumes a constant strike and dip (§5.3). Over the ~6.5 km separating the two outcrops along strike, small discrepancies between the modeled and true strike and dip could produce significant variation in the stratigraphic heights.

It is possible that this northern package is instead the MMA. That would imply that the fine-grained green gabbronorite is not unique to the UMA but also occurs beneath the MMA. It would not change the argument below: either way, the northern correlatives are magnetic mottled anorthosites with associated pyroxenites, and they lie at heights close to those of the southern packages.

The anorthosite mapped by van der Merwe (2007) and Latypov et al. (2022)

Table 5.1: Lower Main Zone anorthosite outcrops compared across the Draaikraal Lineament. The roadcut of the UMA South of the lineament in particular is well documented and agreed to be a lower Main Zone mottled anorthosite (Setera and VanTongeren, 2018; Latypov et al., 2022). The outcrops north of the lineament as correlated by Latypov et al. (2022) and van der Merwe (2007) are compared with those in this work. Heights are stratigraphic heights above the Merensky Reef (§5.3).

Outcrop	Photos	Height (m)	Thick.	Mag.	Texture and associated lithologies
Upper Mottled Anorthosite (UMA)					
Roadcut UMA, South −25.2274, 30.05494	49, 53, 54	564 m	∼0.5 m	yes	Anorthosite with ∼2cm oikocrystic pyroxene mottles overlying ∼0.5 m coarse feldspathic pyroxenite with anorthositic blocks, over fine-grained green gabbronorite with ∼ 1cm blebby bottle-green CPX; irregular, non-planar contacts
"UMA" of Latypov et al. (2022), North −25.1767, 30.07222	86	1030 m	∼10 cm	no	Anorthosite with <1 cm equant pyroxenes. No other unusual lithologies observed. Planar, concordant contacts
"UMA" of this work, "LMA" of Latypov et al. (2022), North −25.1821, 30.09749	90	398 m	>10 cm	yes	Anorthosite with coarse oikocrystic pyroxene mottles over coarse pyroxenite
Middle Mottled Anorthosite (MMA)					
Valley outcrop MMA, South −25.227, 30.06419	103–109	373 m	> 2 m	yes	Very thick; layers of oikocrystic mottled anorthosite and pyroxenite with both planar and wavy, discordant contacts
"MMA" of Latypov et al. (2022), North −25.1827, 30.08279	87	708 m	not measured	no	Anorthosite with pyroxene spots and foliated (sheared) pyroxenes, no pyroxenites.
"MMA" of this work, North **[TODO: coordinates]**	115	333 m	> 4 m	yes	Thick package of anorthosites with coarse oikocrystic pyroxene mottles and pyroxenites with sharp lower contacts that grade upward to anorthosite in ∼ 20 cm layers; cleavage discordant with modal layering

as the UMA north of the Draaikraal Lineament (photo 86) is very small (~10 cm), non-magnetic, and lacks the associated coarse pyroxenite and fine-green gabbronorite lithologies; it does not resemble the UMA outcrop on the Roossenekal-Lydenburg road. Furthermore, it is significantly stratigraphically higher: ~1030 m above the Merensky Reef. van der Merwe (2007) does not report displacement of the UG2 over the Draaikraal Lineament, so it is interesting that their UMA is given ~470 m of stratigraphic displacement.

Similarly, the discordant MMA correlation they offer does not stand up. The MMA is supposed to be a striking and very large marker in the lower Main Zone (photos 103–109), and yet the outcrop they correlate (photo 87) is a small exposure with subtle, non-magnetic pyroxene spots/mottles, and no associated blebby or coarsely layered pyroxenites. I instead find the concordant correlation of a very large package of magnetic mottled anorthosites with coarsely layered pyroxenites at photo 115 much more plausible.

Neither correlation places the anorthosites at exactly the same height on both sides of the lineament, and neither needs to. The thickness of the anorthosite packages and of the layers between them may vary along strike, shifting their relative positions coherently even without invoking a fault. Nonetheless, I prefer the simplest explanation that explains the data. My correlation follows petrography, and happens to require the smallest lateral change to explain the observations: the mottled anorthosites can be followed across the Draaikraal Lineament without invoking a fault.

Because there are essentially no detailed studies of the lower Main Zone anorthosites (similar to Mitchell and Manthree, 2002, for the GMA of the transitional unit), it is easy to fall prey to one's confirmation bias: one can find an anorthosite in the lower Main Zone without too much trouble. Detailed and careful petrography of these marker beds is necessary to make correct correlations. The strategy of Latypov et al. (2022) to use these marker beds to understand the structure and petrogenesis of the RLS is excellent; however, I would argue instead that one should first focus on one particularly remarkable anorthosite package (perhaps the MMA or UMA), and should follow it along strike. For this field area, I would want to do a detailed characterization of the Transitional unit, LMA, MMA, and UMA stratigraphy to the north of the Draaikraal Lineament to get complete sections of the remarkable, but incomplete, outcrops exposed along the Roossenekal-Lydenburg road. Then

I would follow the anorthosites up north to examine how their strike might change, as their regionally concordant NW dip would quickly become discordant to the north. I consider the following open questions to remain:

1) Are the LMA, MMA, and UMA generally planar features, irregular contacts aside?

2) For how long are they laterally continuous?

3) Where and why are they discordant to the regional layering, if ever?

4) Do they abut the floor?

5) How do the distances between them vary at different locations within the intrusion?

Origin of the Mottled Anorthosites in the Lower Main Zone

Surprisingly little attention has been given to the spectacular anorthositic sequences of the lower Main Zone. More often, anorthosites from the Critical (or Transitional) Zone are studied (Mitchell and Manthree, 2002; Maier et al., 2016; Mitchell et al., 2019; Maier et al., 2020; S. J. Barnes et al., 2021), or at the top of the Main Zone at the Pyroxenite Marker (Bourdeau et al., 2021; Bourdeau et al., 2023). The Upper Mottled Anorthosite (UMA) and Pyroxenite Marker of the Main Zone are known to demarcate significant geochemical and isotopic changes in the RLS stratigraphy (Klemm et al., 1985; F. J. Kruger, 1994; VanTongeren and Mathez, 2013), most often ascribed to magma recharge (§B.4, "Radiogenic Isotope Stratigraphy of the RLS").

In the East and West Limbs, the mottles of the lower Main Zone mottled anorthosites (LMA, MMA, UMA) are surprisingly iron-rich– the pyroxenes have much lower Mg# than the surrounding gabbronorites (Scoon and Mitchell, 1994; Mitchell et al., 1998). Though I was surprised to find the mottles of these three units were magnetic at first, it had already been known that they produce subtle but distinct magnetic anomalies observable in aerial surveys, at least within the West Limb (Viljoen and Scoon, 1985; Cole et al., 2013). Viljoen and Scoon (1985) point out that the stratigraphic magnetic anomalies of the mottled anorthosites of the lower Main Zone are not present near structural weaknesses, like faults.

Scoon and Mitchell (1994) make the provocative argument that the silicate iron-rich ultramafic pegmatites (IRUPs) of the upper Critical and lower Main

Zones are sourced from accumulated evolved fluids interstitial to these mottled anorthosites of the lower Main Zone. These putative residual melts are argued to flow laterally, eventually exploiting structural weaknesses to percolate downward to replace plagioclase-rich, magnesian cumulates with olivine-pyroxene Iron-Rich Ultramafic Pegmatites (IRUP), a lithology where plagioclase is conspicuously absent. This bold argument seems unnecessary. It is easy to explain, as Scoon and Mitchell (1994) also do, that IRUPs likely originated from downward-percolating immiscible Fe-oxide melts from the Upper Zone. Strontium isotopes within IRUPs are uniform regardless of stratigraphic height and are identical to the Upper Zone (Scoon and Mitchell, 1994). IRUPs are known to occupy any location along the stratigraphic column between the Upper Zone and their lowest occurrences in the upper Critical Zone, and form a geochemical continuum that records them becoming more magnesian and silica-rich with depth, transitioning from Fe-Ti oxide pegmatites to Fe-rich silicate pegmatites, telling a clear story of evolution/hybridization of downward-percolating, dense Fe-rich liquids.

Still, the strontium isotopes of these mottled anorthosites and some stratigraphically proximal gabbronorites are often discontinuous with the otherwise homogeneous, isotopically evolved Main Zone immediately surrounding them, and instead are, remarkably, identical to the Upper Zone and IRUPs (Karykowski et al., 2017; F. J. Kruger, 1994). It may very well be true that the IRUPs are related to the mottled anorthosites of the lower Main Zone, though I would instead propose that these anorthosites might have served as porous "aquifers" that trapped IRUP melts originating from the Upper Zone, rather than pose the mottled anorthosites as the source of these iron-rich melts.

Additionally, compelling geochemical arguments have been presented on texturally similar mottled anorthosites in the Critical Zone that argue for formation by reactive porous flow of a hydrothermal fluid including the presence of evolved but Ni–rich olivine, hydrous phases and sulfides, and extreme reverse zoning of plagioclase surrounding mottles (Maier et al., 2020, see also §B.5, "Origin of Enigmatic Monomineralic Igneous Rocks"). The presence of water is known to favor olivine crystallization at the expense of pyroxenes, as well as favoring more anorthitic plagioclase over albite (Baker and A. E. Boudreau, 2019).

I envisage mottled, stratiform anorthosites to form where low-permeability structural traps collect upward-percolating reactive hydrothermal fluid that

locally favors plagioclase crystallization and pyroxene dissolution. EPMA work on the mottles of the LMA, MMA, and UMA would bring clarity to the relative importance of evolved interstitial fluids, potential IRUP contaminants, and hydrothermal fluids on the formation of mottled anorthosites in the lower Main Zone and to stratiform anorthosites in general.

Rethinking the Marginal Zone: Arguments for Incremental Assembly The Marginal Zone has been traditionally understood to be the chilled margin of the RLS (Harmer and Sharpe, 1985). However, detailed work reveals that the lowest stratigraphy of the RLS is complex. The Marginal Zone is not always present at the floor of the RLS, especially to the southeast (Molyneux, 2008; van der Merwe, 2007; Latypov et al., 2020). Marginal Zone norites may locally intrude the Lower Zone (A. H. Wilson et al., 2017; Yudovskaya et al., 2021), and the Marginal Zone overlies an additional Bushveld-related ultramafic sequence referred to as the Apiesdooringdrai peridotite body or Basal Ultramafic Sequence (BUS) at Burgersfort that is more likely to contain chilled components of the earliest RLS-related intrusions (A. H. Wilson, 2012; A. H. Wilson et al., 2017). The BUS, Marginal Zone, and Lower Zone all contain reacted sedimentary "rafts" or xenoliths with igneous mineralogy (Yudovskaya et al., 2021). Thinner ultramafic and mafic sills likely related to Bushveld magmatism intrude the floor rocks kilometers below the mapped Marginal Zone (Sharpe and Hulbert, 1985). In my field work, I observed that Marginal Zone sills seemed to concordantly intrude the Transvaal sediments below the RLS for many kilometers.

The lack of a cohesive igneous stratigraphy and the geochemical and isotopic variability of the Marginal and Lower Zones led Mitchell and Scoon (2023) and Scoon and Mitchell (2023) to propose that the Marginal and Lower Zones were assembled incrementally by magmas of different compositions (see also §B.5, "Spatial and Temporal Scales of Magma Chambers in the Crust, Pluton Assembly Mechanisms"). I would extend their arguments to the entire RLS. I posit that a series of stacked peridotites, pyroxenites, norites, gabbronorites, and gabbros were emplaced, somewhat self-sorted according to density and the compositional control on stalling depth (more primitive magmas will crystallize at higher temperatures) at the location of the BUS, RLS and Marginal Zones presently. I suggest that postcumulus modification by reactive porous flow of late stage magmas and/or hydrothermal fluids allowed for

dissolution and recrystallization to produce phaneritic textures within the RLS, and that these fluids interacted with the Marginal Zone to a far lesser extent than the greater RLS. These fluids reequilibrated and reacted with the RLS within a depth-controlled thermal gradient which produced the RLS-scale cohesive geochemical trends that are mostly uncorrelated with isotopic and lithological boundaries (see Eales et al., 1986). Subconcordant intrusion of magmas as sills into the country rock results in the inclusion of abundant but comparatively volumetrically small strands of sedimentary rock in the Lower, Marginal, Critical, and Main Zones of the RLS. These strands react to igneous mineralogy by reactive porous flow along grain boundaries and develop planar modal layering due to lateral transit of fluids and evolved magmas sourced from the Thabazimbi-Murchison Lineament (TML) and they serve as traps for hydrothermal fluids that locally supported the reaction, dissolution, and crystallization of mafic lithologies into mottled anorthosites below. The Merensky Reef represents a structural trap for a reduced supercritical carbon-based fluid (Ballhaus and Stumpfl, 1985, see also §B.4, "Unusual Volatile-bearing Phases in the Lower RLS"). The isotopic homogeneity of the Upper Zone (§B.5, "Open or Closed System") allows for a more orthomagmatic interpretation: perhaps it represents a magma chamber where evolved tholeiitic magmas and hydrothermal fluids collected, and magnetite/anorthosite layers were crystallized by cycles of depressurization, cooling, and perhaps magma recharge.

Significance of Rock Cleavage and Modal Layering Orientation in the RLS

The discordance of rock cleavage and modal layering (§5.4; Figure 5.4) suggests that cleavage and modal layering are formed by distinct processes that may have occurred at different times, with modal layering likely predating the formation of rock-cleavage.

There may have been multiple processes generating superimposed fabrics in the plutonic RLS. For instance, rock fabrics may have been developed in two phases, perhaps a primary fabric (modal layering) developed by gravity and crystal settling (§B.1), then a second by lateral flow of hydrothermal fluids, or by differential contractions during cooling. Care must be taken when interpreting planar measurements in plutonic rocks of the RLS. The bimodal planes of the RLS in this field area may be due to an initial development of south-

dipping igneous fabric above the Kliprivier Synform as magmas filled a local basin towards the south before another generation of paleohorizontal fabric developed perpendicular to thermal gradients due to differential contraction during cooling.

5.6 Conclusions

1) The field data of this study support the interpretation that the Kliprivier Synform is a monocline synmagmatic with the Critical Zone below UG2, predating the UCZ at UG2 and up. There is no displacement of marker beds above UG2, including all mottled anorthosites. The country rocks and Marginal Zone sills of the Kliprivier Synform are roof rocks to the Critical Zone below UG2 and are the floor to undeformed Transitional and Main Zone.

2) The Draaikraal Lineament is best explained as the surface projection of the subvertical axial plane of the Kliprivier Synform. It is hypothesized that trapped fluids from the emplacement of the underlying Critical Zone may have exploited the deformed Kliprivier Synform, developing a vertical, planar fluid conduit at the Draaikraal Lineament where upwardly migrating fluids may have escaped, developing recessive lithologies. The regional recessive topography of the Main Zone south of the Draaikraal Lineament is proposed to originate from relatively high degrees of contamination from devolatilized/melted underlying sediments relative to the Main Zone lithologies at the same stratigraphic height to the north of the Draaikraal Lineament.

3) The discordant mottled anorthosite correlation above and below the Draaikraal Lineament by van der Merwe (2007) and Latypov et al. (2022) is locally incorrect. The field work presented here instead suggests that the mottled anorthosite marker beds of the lower Main Zone are concordant and undisplaced by the Draaikraal Lineament. More careful, detailed field work will be necessary to ensure correct correlation of the many understudied anorthosites of the lower Main Zone.

4) Pyroxenite aggregate tennisballs are found in fine-grained Marginal Zone Gabbros. This supports the interpretation that the coarse-grained porphyritic gabbronorite host of the TBM at Middelkraal is a member of the Marginal Zone after Mitchell and Scoon (2023). It is difficult to explain

the development of a diapiric tail on the nascent pyroxene tennis balls in the fine-grained Marginal Zone with the model of Khoza (2021), although this observation allows the models of Mitchell and Scoon (2023), Smith et al. (2023), and Lee and Sharpe (1979).

5.7 Acknowledgment

Profound thanks are due to Professor James Roberts for his charismatic introduction to the plethora of all-consuming geological problems of "The Beast" (the Bushveld Complex). He is also owed thanks for warmly hosting me as a visitor to the University of Pretoria in 2023, for providing resources to explore the Draaikraal area, for accompanying me for several days of field work in Draaikraal, and for teaching me alongside UP undergraduates on a field course of the wondrous Bushveld.

Appendix A

DRAAIKRAAL FIELD OBSERVATION DATA

A.1 Field Observations

Every geotagged field observation of the July 2023 campaign is listed in Table A.1, in order of stratigraphic height relative to the Merensky Reef ($h = 0$ m), with location, interpreted map unit and lithology, structural readings, and the IDs of the photographs taken there. Detailed field notes are footnotes to the observation names.

Table A.1: Field observations of the Draaikraal campaign (July 2023). Lat./Lon.: WGS84, decimal degrees. Elev.: elevation (m). Unit: final interpreted map unit and Lith.: lithology code (§5.4). Structure: bedding (Bed.), cleavage (Clv.), contact (Cont.) or modal layering (M.L.), as strike/dip with strike an azimuth from north. Photos: field photograph numbers; those reproduced in §A.2 or as figures are links, the others are in the supplementary photo album. Field descriptions and notes are footnotes to the observation name, at the foot of the page. '–': not recorded.

Name	Ht.	Lat.	Lon.	Elev.	Unit	Lith.	Structure	Photos
RZ22	-933	-25.24760	30.11812	2146	Rz	Msp	–	–
QZ27	-724	-25.24040	30.10576	2076	Rz	Qz	–	–
QZ7	-706	-25.23020	30.11221	1958	Rz	Qz	–	–
RZ54	-700	-25.23980	30.10496	2044	Rz	Msp	–	–
CONT MARG E Q W?[1]	-697	-25.22880	30.11288	1960	Rz	Qz	–	–
D3NB	-683	-25.22540	30.11481	1960	Rz	Qz	Bed. 145/30	–
QZ1	-676	-25.22530	30.11455	1956	Rz	Qz	–	–
D3NA[2]	-674	-25.22590	30.11390	1929	Rz	Qz	Bed. 120/30	–
RZ43	-669	-25.23920	30.10392	2015	Rz	Msp	–	–
D7NC[3]	-661	-25.23900	30.10369	2010	Rz	Qz	–	–
RZ8	-654	-25.22500	30.11369	1945	Rz	Msp	–	–
D3NC[4]	-649	-25.22510	30.11337	1940	Rz	QzMsp	–	34
RZ21	-644	-25.22500	30.11326	1947	Rz	Msp	–	–
RZ24	-643	-25.23900	30.10279	1997	Rz	Msp	–	–
RZ36	-642	-25.23860	30.10304	1995	Rz	Msp	–	–
QZ112	-639	-25.23860	30.10289	1995	Rz	Qz	–	–
D3SF[5]	-638	-25.22610	30.11207	1929	Rz	ScP	–	37, 38

continued on next page

[1] Quartzite. Quartzite follows topo line locally here.
[2] Compass trouble, resolved.
[3] Alternating quartzite-marginal zone photo, not very good one.
[4] No faulting; the quartz and marginal zone is interbedded.
[5] Qz-Px Pegmatite. Hosted in migmatitic quartzite. Schist banding deformed around pegmatite.

Table A.1 (continued)

Name	Ht.	Lat.	Lon.	Elev.	Unit	Lith.	Structure	Photos
SCHIS	-636	-25.22580	30.11224	1937	Rz	Sc	–	–
QZ2	-631	-25.22510	30.11260	1950	Rz	Qz	–	–
QZ3	-631	-25.22460	30.11297	1954	Rz	Qz	–	–
RZ11	-629	-25.22490	30.11266	1952	Rz	Msp	–	–
QZ6	-629	-25.22520	30.11238	1946	Rz	Qz	–	–
QZ5	-626	-25.22480	30.11255	1950	Rz	Qz	–	–
RZ31	-625	-25.22600	30.11153	1926	Rz	Msp	–	–
RZ33	-622	-25.22460	30.11252	1946	Rz	Msp	–	–
D3SE[6]	-621	-25.22460	30.11251	1950	Rz	Msp	–	36, 40
D3SD[7]	-619	-25.22450	30.11247	1946	Rz	ScQz	–	35
RZ71	-617	-25.22690	30.11044	1911	Rz	Msp	–	–
SCHIST	-615	-25.22440	30.11236	1944	Rz	Sc	–	–
QZ4	-615	-25.22430	30.11241	1945	Rz	Qz	–	–
RZ41	-614	-25.22410	30.11254	1948	Rz	Msp	–	–
RZ110	-614	-25.23840	30.10183	1976	Rz	Msp	–	–
RZ51	-614	-25.22290	30.11342	1945	Rz	Msp	–	–
RZ61	-599	-25.22550	30.11068	1907	Rz	Msp	–	–
RZ32	-588	-25.22590	30.10982	1900	Rz	Msp	–	–
QZ26	-575	-25.23760	30.10054	1948	Rz	Qz	–	–
D7SE[8]	-569	-25.23690	30.10081	1946	Rz	ScP	–	70, 71
SHIST	-569	-25.23690	30.10078	1947	Rz	ScP	–	–
RZUP	-565	-25.23740	30.10019	1935	Rz	ScPMsp	–	–
D7SD[9]	-559	-25.23740	30.09991	1936	Rz	ScP	–	69
D2NC[10]	-555	-25.22110	30.11222	1993	Rz	QzMsp	–	14, 23, 28
D7NA[11]	-552	-25.23700	30.09985	1924	Rz	QzMsp	Bed. 220/25	67
SCH1	-551	-25.23700	30.09984	1930	Rz	ScP	–	–
RZMAG	-549	-25.23690	30.09979	1919	Rz	Msp	–	–
RZ	-532	-25.22390	30.10878	1909	Rz	Msp	–	–
D1ND[12]	-507	-25.23730	30.09753	1926	Rz	Msp	–	6
D1NC[13]	-500	-25.23720	30.09730	1924	Rz	Msp	–	–
RZ3	-485	-25.22250	30.10765	1910	Rz	Msp	–	–
RLS	-481	-25.22350	30.10665	1890	Cz	Pg	–	–

continued on next page

[6]Fine grained, light rock. Dacite? Silicified. Pockmarks weathering proud in paleovertical pipes.

[7]Migmatized quartzite. Banded. Light bands of quartz preserve original sedimentary texture. Grains rounded and 2 mm. Some bands pink (feldspar) some green (chlorite, epidote?)

[8]Silicate Pegmatite. within migmatized qzite very close to magnesite skarn. Myrmekitic quartz-K-spar, huge OPX, small vibrant rare green and blue crystals: apatite, CPX, beryl?

[9]Magnesite Skarn. within migmatized qzite. In ground, hard to tell if dike or concordant with schist host.

[10]Contact. On this ridge, there are multiple bands of quartzites with Marginal Zone between them. Is the marginal zone a sill between layers of quartzite, or is there a fault? (Update, I think it's a strike-slip(?) fault striking E-W dipping to the South with a component of left-lateral shear along the ridge)

[11]Fine marginal gabbronorite with shearing pegmatite veins. sheared pegmatite veins in pockets/pipes orthogonal to bedding, paleo-up. This layer is just above migmatized quartzite and just below fine magnetic marginal zone. Represents fluid transport from migmatized sedimentary rocks, origin of resistance banding? Strike and Dip are for migmatized Qz.

[12]Bladed iridescent white crystals (Plagioclase) in 2 cm thick planar features intersect orthogonally in outcrop. The plagioclase planes are surrounded by 10 cm zone of oikocrystic augite. The southwestern limb pinches in and out and occasionally overlaps itself in parallel, a la boudinage.

[13]Note that all marginal zone rocks are characterized by dark resistance bands, perhaps cataclastic, reflecting warm-brittle deformation with finely recrystallized zones. They stand proud. They frequently strike E-W, secondarily, NE-SW, tertiarily. NW-SE, but can vary and appear pretty random. This feature is diagnostic of the marginal zone.

Table A.1 (continued)

Name	Ht.	Lat.	Lon.	Elev.	Unit	Lith.	Structure	Photos
RZ1	-481	-25.22030	30.10931	1973	Rz	Msp	–	–
D1Sb[14]	-479	-25.23690	30.09644	1891	Rz	Msp	–	19
D1A[15]	-479	-25.23690	30.09644	1891	Rz	MspTB	–	4
MZ3	-478	-25.22290	30.10699	1902	Cz	Pg	–	–
D2SB[16]	-477	-25.21950	30.10983	2004	Rz	Qz	–	–
RZ2	-470	-25.22270	30.10679	1901	Rz	Msp	–	–
D2SH[17]	-469	-25.22070	30.10839	1950	Rz	ScP	–	29, 30
RQCONT	-465	-25.21920	30.10955	2014	Rz	QzMsp	–	–
RZ20	-456	-25.23540	30.09654	1903	Rz	Msp	–	–
RZ5	-441	-25.23760	30.09423	1918	Rz	Msp	–	–
D2ND[18]	-440	-25.21820	30.10918	2020	Rz	Qz	Bed. 175/25	–
D11SA[19]	-440	-25.23770	30.09405	1899	Rz	Msp	–	–
QZ	-433	-25.21900	30.10814	1990	Rz	Qz	–	–
FOOLLOW TOPO								
RZ4	-431	-25.23700	30.09419	1906	Rz	Msp	–	–
RZ6	-430	-25.23860	30.09298	1923	Rz	Msp	–	–
D2NE[20]	-426	-25.21670	30.10968	2031	Rz	Qz	Bed. 225/25	–
D4NB[21]	-422	-25.24150	30.09048	1932	Rz	QzMsp	Bed. 240/25	52
QZ51	-417	-25.21880	30.10754	1993	Rz	Qz	–	–
D10NK	-415	-25.24210	30.08966	1917	Rz	Qz	Bed. 235/25	–
RZ7	-413	-25.23870	30.09213	1915	Rz	Msp	–	–
D4SA[22]	-412	-25.24090	30.09036	1895	Rz	QzMsp	–	50, 51
SCHI	-403	-25.24040	30.09025	1882	Rz	Sc	–	–
QZ61	-402	-25.21890	30.10676	1980	Rz	Qz	–	–
QZ41	-402	-25.21870	30.10691	1987	Rz	Qz	–	–
QZ	-395	-25.21680	30.10808	2001	Rz	Qz	–	–
QZ29	-393	-25.24210	30.08861	1907	Rz	Qz	–	–
QZ71	-391	-25.21900	30.10613	1975	Rz	Qz	–	–
RZ28	-388	-25.24200	30.08845	1906	Rz	Msp	–	–
SCH	-387	-25.24010	30.08972	1870	Rz	Sc	–	–
RZ12	-381	-25.23990	30.08955	1866	Rz	Msp	–	–
D2SA[23]	-380	-25.23000	30.09696	1860	Cz	PgTB	–	–

continued on next page

[14]Common Marginal zone Medium-fine grained gabbronorite.

[15]Fine-grained Magnetite Gabbro with gabbronoritic blocks. Fine-grained magnetite gabbro includes rounded, elongate blocks of oikocrystic gabbronorite preferentially horizontally imbricated, ∼6-10 cm along axis. Also a bunch of rounded gabbronorite inclusions that may represent the spherical aggregates of the TBM.

[16]Sampled because quartzite here all have iron stained interiors. The weathered exteriors are clear. Quartzites served as a conduit for mobilized iron-rich fluids?

[17]Schist with Skarn. Swirled and deformed schist with marginal zone, with thin clear banding. Non-magnetic. Xenolith? Associated with skarn pegmatite veins: 60% Feldspar, 40% amphibole? Big very dark crystals that I think I see 120 degree cleavage on... Likely devolatilizing xenolith.

[18]Measurable bedding planes at top of ridge: Strike: 265, Dip: 25. Also, One bedding plane preserved lovely sedimentary ripples striking E-W.

[19]Coarse grained gabbronorite. Is it critical-margin zone? Are resistance bands chromite bearing?

[20]Different strike, same dip. Strike: 315, dip: 25. Quartzites are rotated around the top of the ridge. NW-SE vertical fault or E-W South dipping fault can neatly explain rotation.

[21]Contact between floor quartzite and marginal zone. Bed plane of quartzite measured.

[22]Contact between marginal zone and migmatized quartzite. Migmatitic quartzite ∼10m thick. At contact is non-magnetic leucocratic gneiss, then fine magnetic marginal zone: 10cm, then coarse grained marginal zone: 30 cm, then common fine marginal zone.

[23]Common Main Zone Coarse Grained Gabbronorite (Office interp: Leucogabbronorite; porphyritic gabbronorite) On Dirt road to Kliprivier Farms, Boulder piles outcropping of main zone.

Table A.1 (continued)

Name	Ht.	Lat.	Lon.	Elev.	Unit	Lith.	Structure	Photos
RZ35	-380	-25.21890	30.10566	1965	Rz	Msp	–	–
D2SF[24]	-379	-25.21550	30.10829	1994	Rz	Msp	–	24
RZ42	-379	-25.21940	30.10519	1950	Rz	Msp	–	–
QZ8	-377	-25.21920	30.10529	1954	Rz	Qz	–	–
RZ9	-370	-25.23920	30.08952	1850	Rz	Msp	–	–
RAT	-368	-25.23630	30.09174	1883	Rz	Msp	–	–
RZ511	-368	-25.21970	30.10442	1935	Rz	Msp	–	–
RZLQZR	-366	-25.21960	30.10440	1938	Rz	QzMsp	–	–
RZ53	-362	-25.21930	30.10448	1945	Rz	Msp	–	–
RZ113	-359	-25.24170	30.08722	1876	Rz	Msp	–	–
IS1	-358	-25.23030	30.09577	1868	IS	IS	–	–
RZ62	-355	-25.21960	30.10390	1934	Rz	Msp	–	–
D2SG[25]	-352	-25.21390	30.10811	1956	Cz	Dcg	–	27
QZ9	-352	-25.22000	30.10342	1929	Rz	Qz	–	–
SCI	-350	-25.23900	30.08890	1888	Rz	Sc	–	–
D9NE	-348	-25.25130	30.07965	1929	Rz	Qz	Bed. 180/15	–
QZ10	-347	-25.22000	30.10320	1928	Rz	Qz	–	–
SCH11	-343	-25.24070	30.08725	1881	Rz	Sc	–	–
RZ27	-341	-25.24060	30.08725	1882	Rz	Msp	–	–
D9SD[26]	-340	-25.25150	30.07907	1912	Rz	Msp	–	95
QZ11	-339	-25.21970	30.10308	1938	Rz	Qz	–	–
SCH2	-338	-25.23940	30.08802	1887	Rz	Sc	–	–
RZ82	-338	-25.23960	30.08784	1886	Rz	Msp	–	–
D9NC[27]	-335	-25.25180	30.07858	1908	Rz	Msp	–	–
QZ12	-334	-25.21970	30.10285	1939	Rz	Qz	–	–
D9NF[28]	-333	-25.25050	30.07951	1918	Rz	MspTB	–	96
RZ23	-331	-25.21720	30.10466	1957	Rz	Msp	–	–
RZ72	-330	-25.21940	30.10291	1940	Rz	Msp	–	–
RZ81	-318	-25.21920	30.10248	1940	Rz	Msp	–	–
RZ19	-316	-25.21700	30.10410	1948	Rz	Msp	–	–
RZ25	-312	-25.24970	30.07913	1910	Rz	Msp	–	–
D9NG[29]	-312	-25.24980	30.07904	1910	Rz	Msp	–	97, 98
D5NC[30]	-299	-25.21880	30.10195	1946	Rz	Msp	Clv. 145/25	57
RZ44	-290	-25.24030	30.08497	1844	Rz	Msp	–	–
RZ18	-286	-25.21730	30.10245	1940	Rz	Msp	–	–
RZ91	-280	-25.21900	30.10086	1928	Rz	Msp	–	–
QZ31	-264	-25.21640	30.10209	1933	Rz	Qz	–	–
QZ28	-252	-25.24750	30.07796	1893	Rz	Qz	–	–

continued on next page

These rocks are the building materials for much of the structures in the area. Weathering is quite distinct from Marginal zone. Both have exfoliation/onion skin weathering. Main Zone is prouder, forming impressive rounded boulders like a granite. Weathered faces have distinct dalmatian pattern. A large boulder (float) at this outcrop contains a characteristic pyroxene "tennisball", a well-known marker at the base of the Main Zone.

[24]Coarse Grained Magnetite Gabbro. Actually I am now thinking this sample is similar to the magnetite gabbro D1Sa, but coarser grained. That sample too had zones of coarse grains, and this one also had blocks of oikocrystic augite.

[25]Norite. Weathering more closely resembles main zone than nearby marginal zone, but this rock does not look like main zone to me. Non-magnetic.

[26]Fine Grained Marginal Norite. Sampled for prominent resistance band for later analyses.

[27]Chloritized Norite. Plag, OPX, abundant phlogopite, alteration grade lessens up section.

[28]Fine Grained Marginal norite. Droplet forms of coarse pyroxenite xenoliths, pegmatite inclusions?

[29]Paleovertical Pegmatite Dike Photograph, Marginal zone coarsens upsection towards quartzite.

[30]Cleavage.

Table A.1 (continued)

Name	Ht.	Lat.	Lon.	Elev.	Unit	Lith.	Structure	Photos
RZ10	-252	-25.21880	30.09972	1929	Rz	Msp	–	–
QZ13	-248	-25.21950	30.09891	1907	Rz	Qz	–	–
D5SB[31]	-248	-25.21430	30.10287	1919	Cz	A	–	–
D9NH	-244	-25.24740	30.07764	1895	Rz	Qz	Bed. 230/35	–
QZ23	-242	-25.21610	30.10128	1932	Rz	Qz	–	–
QZ14	-241	-25.21960	30.09852	1907	Rz	Qz	–	–
QZ15	-233	-25.21970	30.09807	1907	Rz	Qz	–	–
RPEG1	-229	-25.21390	30.10229	1907	Cz	Dcg	–	–
RZ111	-223	-25.21950	30.09777	1902	Rz	Msp	–	–
D7NF[32]	-218	-25.21350	30.10206	1906	Cz	Dcg	Bed. 180/10	73
QZ110	-215	-25.21610	30.10002	1923	Rz	Qz	–	–
QZ113	-214	-25.24740	30.07621	1874	Rz	Qz	–	–
RZ121	-213	-25.21890	30.09779	1911	Rz	Msp	–	–
QZ16	-213	-25.21910	30.09763	1908	Rz	Qz	–	–
QZ17	-211	-25.21880	30.09778	1915	Rz	Qz	–	–
RZ13	-209	-25.21860	30.09780	1909	Rz	Msp	–	–
QZ18	-205	-25.21850	30.09772	1914	Rz	Qz	–	–
D9NI[33]	-204	-25.24670	30.07624	1865	Rz	ScMsp	–	99, 100, 101, 102
QZ19	-200	-25.21860	30.09741	1912	Rz	Qz	–	–
RZ112	-199	-25.24650	30.07613	1865	Rz	Msp	–	–
D7NG[34]	-197	-25.21220	30.10208	1905	Cz	Dcg	Clv. 230/15	72, 73, 74, 75
SCHI1	-193	-25.24620	30.07609	1863	Rz	Sc	–	–
RZQZLEFT	-193	-25.21920	30.09654	1887	Rz	QzMsp	–	–
QZ20	-187	-25.21880	30.09663	1901	Rz	Qz	–	–
RZ14	-185	-25.21870	30.09657	1900	Rz	Msp	–	–
RZ15	-175	-25.21840	30.09636	1895	Rz	Msp	–	–
RZ26	-173	-25.24550	30.07565	1856	Rz	Msp	–	–
MZ11	-166	-25.22800	30.08851	1837	Cz	Pg	–	–
D5SA[35]	-166	-25.21540	30.09820	1903	Cz	UD	–	–
D6SL[36]	-165	-25.21490	30.09851	1892	Cz	Pg	–	–
RZ16	-162	-25.21870	30.09548	1887	Rz	Msp	–	–
QZ22	-161	-25.21810	30.09591	1898	Rz	Qz	–	–
D5NE	-160	-25.21870	30.09537	1883	Rz	Msp	Clv. 125/25	–
RZ17	-154	-25.21950	30.09445	1875	Rz	Msp	–	–
QZ21	-153	-25.21950	30.09444	1875	Rz	Qz	–	–
QZ25	-147	-25.21770	30.09553	1877	Rz	Qz	–	–
D6SG[37]	-146	-25.21840	30.09491	1872	Rz	QzMsp	–	60

continued on next page

[31] Amphibole Gabbro. Weakly magnetic, amphibole bearing? Rock clearly rotten.

[32] Pegmatitic gabbronorite. large (5 cm) oikocrystic pyroxene.

[33] Schist (Pebble marker?) well mixed clasts of schists, zones of pink felsic melt.

[34] Pegmatitic gabbronorite. Here magnetic, slightly upsection (∼2 m) rocks non-magnetic. Large (30 cm) blobs of pyroxenite rimmed by anorthosite. Pyroxene/Anorthosite ranging from 1/4 to 4. Included blocks with preserve resistance banding.

[35] Thought it looked like the "dark facies" main zone in the field,. Compare to D25g (porphyritic gabbronorite), Quartzite in float (alluvial)

[36] Porphyritic gabbronorite. In notes, described as main zone porphyritic gabbronorite. However these rocks are structurally beneath chrome layers. Closer inspection reveals that pyroxenes are often oikocrystic and large... Incompletely formed Mottled Anorthosite?

[37] Magnetite pegmatite. Host rock magnetic. Pegmatite weathers proud, cutting the rock paleo-up, "sandy" texture in host rock.

Table A.1 (continued)

Name	Ht.	Lat.	Lon.	Elev.	Unit	Lith.	Structure	Photos
D6NK[38]	-145	-25.21500	30.09749	1888	Cz	Pg	Clv. 120/5	–
D7SH[39]	-144	-25.21020	30.10109	1898	Cz	Px	–	78
D6SH[40]	-144	-25.21770	30.09534	1871	Cz	Dcg	–	62
D6SI[41]	-144	-25.21770	30.09534	1871	Cz	Dcg	–	61
QZ111	-140	-25.21760	30.09526	1869	Rz	Qz	–	–
QZ24	-139	-25.21800	30.09487	1869	Rz	Qz	–	–
RPEQ	-135	-25.21480	30.09718	1883	Cz	Dcg	–	–
PEG	-133	-25.21140	30.09968	1893	Cz	Dcg	–	–
D6NJ[42]	-132	-25.21540	30.09655	1880	Cz	Dcg	Clv. 150/10	63
RPEG	-130	-25.21540	30.09644	1877	Cz	Dcg	–	–
RZ34	-124	-25.21910	30.09334	1858	Rz	Msp	–	–
RZ37	-120	-25.24380	30.07447	1840	Rz	Msp	–	–
RZ52	-118	-25.22040	30.09203	1846	Rz	Msp	–	–
D7NI[43]	-100	-25.20893	30.09996	1876	Cz	PgTB	–	79, 119
D7SJ[44]	-100	-25.20890	30.09995	1876	Cz	Pg	–	–
MZPOR	-96	-25.20910	30.09963	1873	Cz	Pg	–	–
CZ3	-94	-25.20880	30.09978	1877	Cz	Ln	–	–
D6NM[45]	-93	-25.21080	30.09820	1868	Cz	Ln	Clv. 233/16	64
MZ5[46]	-87	-25.24180	30.07439	1829	Cz	UD	–	–
D6SN[47]	-86	-25.21040	30.09815	1863	UG2	UG2	–	65, 66
CZ2	-83	-25.21040	30.09804	1865	Cz	Ln	–	–
CZ1	-72	-25.20930	30.09835	1862	Cz	Ln	–	–
D4ND[48]	-72	-25.21690	30.09254	1847	Cz	Pg	Clv. 220/40	–
CZ	-58	-25.20720	30.09933	1884	Cz	Ln	–	–
IS	-52	-25.24350	30.07152	1840	IS	IS	–	–
MZ2[49]	-51	-25.21740	30.09118	1847	Cz	Pg	–	–
CZ5	-48	-25.20630	30.09959	1898	Cz	Ln	–	–
D7SK[50]	-33	-25.20600	30.09920	1919	TU	Ma	–	81
D11SB[51]	162	-25.22810	30.07313	1803	Mz	Pg	–	–
D11SC[52]	207	-25.22700	30.07188	1806	LMA	Ma	–	–
D11NH[53]	333	-25.19250	30.09242	1917	MMA	Ma	M.L. 130/15	115
D5NF	362	-25.22690	30.06477	1803	Mz	UD	Clv. 240/25	–

continued on next page

[38] Porphyritic gabbronorite.

[39] Sulfide-pyroxenite.

[40] Coarse-grained Feldspathic Pyroxenite. Non-magnetic, Floor of critical zone? "Flatreef"?

[41] Coarse-grained Feldspathic Pyroxenite. Coarser grained rocks here are more rotten.

[42] Pegmatite gabbronorite.

[43] Pyroxene tennisballs in Gabbronorite. Structurally below chromite?

[44] Gabbronorite. small amount of interstitial CPX. Transitional zone?

[45] Fine Layered Anorthosite-norite.

[46] No field notes. Assigned Cz because it is stratigraphically below my modeled TU.

[47] Magnetic chromite seam w/ oikocrystic pyroxenite and anorthosite. abundant blocks. chromite hosted with pure anorthosite and in massive layer with oikocrystic pyroxene. Top layers are magnetic (magnetite-bearing?)

[48] Coarse Gabbronorite. Indirect measurement, bedding planes in a marsh.

[49] Assigned Mz because it is undeformed / overlying Kliprivier synform.

[50] Mottled Anorthosite. Non-magnetic. Interpreted as proximal Merensky Footwall based distance from UG2.

[51] Porphyritic Gabbronorite. Rotten. Initially confused for mottled anorthosite.

[52] Mottled Anorthosite. Lower Mottled anorthosite; outcrop source of James' favorite boulder.

[53] Large cliffside mottled anorthosite outcrop. Modal Layering.

Table A.1 (continued)

Name	Ht.	Lat.	Lon.	Elev.	Unit	Lith.	Structure	Photos
D10SJ[54]	373	-25.22700	30.06419	1808	MMA	MaBap	Clv. 220/15	103–110
MZ1	377	-25.22870	30.06271	1797	Mz	Pg	–	–
MA	378	-25.22920	30.06219	1767	MMA	Ma	–	–
DS	379	-25.19100	30.09140	1903	Dl	Dl	–	–
D8SI[55]	398	-25.18210	30.09749	1981	UMA	MaFgog	Clv. 220/20	90
D8SH[56]	413	-25.18210	30.09678	1984	Dl	Dl	Cont. 185/25	89
D11NF[57]	415	-25.19710	30.08507	1896	Mz	Lg	M.L. 130/15	113
MA1	423	-25.22600	30.06264	1805	MMA	Ma	–	–
D10SI[58]	434	-25.22550	30.06250	1801	MMA	MaBap	–	–
D8NJ	446	-25.18260	30.09486	1979	Mz	Hg	Clv. 230/15	–
D11SG[59]	451	-25.19410	30.08572	1911	UMA	Ma	–	114, 117
RESB1ND	454	-25.18230	30.09475	1983	Mz	Hg	–	–
D5SG[60]	509	-25.21960	30.06343	1789	UMA	MaPxFgog	–	–
D3SG	564	-25.22740	30.05494	1780	UMA	MaPxFgog	–	49, 53, 54
(D4NC)[61]								
D11NI[62]	622	-25.19160	30.07954	1865	Mz	Lg	Clv. 240/15	–
MODALLAYER	687	-25.18070	30.08540	2054	Mz	Lg	–	–
D8SG[63]	708	-25.18270	30.08279	2022	Mz	Sa	–	87
MZD	739	-25.23480	30.04130	1790	Mz	Hg	–	–
MODAL	743	-25.19140	30.07401	1828	Mz	Lg	–	–
LAYER								
IS2	865	-25.19950	30.06205	1781	IS	IS	–	–
D11SE[64]	867	-25.18590	30.07260	1886	Mz	Lg	–	112
D11ND[65]	893	-25.18530	30.07186	1891	Mz	SaLg	–	111
D8SC[66]	1030	-25.17670	30.07222	1968	Mz	Sa	M.L. 140/25	86
D8NE	1035	-25.17780	30.07114	1955	Mz	UD	M.L. 130/10	–

continued on next page

[54]Mottled anorthosite with blebby/blocky recrystallized pyroxenite. Big Outcrop, undulating contacts of MA, often dipping east. Sometimes appear discordant, though others appear concordant: Can be explained by thermal erosion, deposition of surface with meter-scale hummocks and swales.

[55]Mottled Anorthosite. Interpreted as LMA. Magnetic mottles. Includes fine-grained gabbronorite with large anhedral CPX phenocrysts (<5 cm)

[56]Dolerite Sill. Strongly magnetic, with well-developed fine blocky cleavage from stress field. Inherited phenocrystic blocky 1 cm plagioclase with strong oscillatory zoning. Likely later intrusive, not necessarily concordant.

[57]Layered/porphyritic gabbronorite. Fine modal layering, near "peace" rock.

[58]Mottled anorthosite with blebby/blocky recrystallized pyroxenite. MMA.

[59]Mottled Anorthosite. Very large magnetic mottles (<10 cm) with large (< 0.5 cm) magnetite.

[60]Mottled Anorthosite. Float, outcrop in sight. Didn't want to trespass, Also found green fine grained rock with large anhedral CPX.

[61]Mottled Anorthosite over strongly graded pyroxenite-anorthosite over fine grained Gabbronorite with large (<5 cm) anhedral CPX. Gave a ride to an African man walking home from work at Kliprivier farms who spoke no English. Stopped by an outcrop and sampled in the dark. Vantongeren maps as Upper Mottled anorthosite.

[62]Finely Layered Gabbronorite. Good cleavage, Mottled Anorthosite Nearby?

[63]Sheared Spotted Anorthosite. Strongly foliated, suggesting co-crystallization shearing. Biggest mottles: ∼3cm x 0.5 cm. Interpreted as MMA by RL et al., 2022.

[64]Finely layered gabbronorite. Mottled anorthosite itself is in float here, but graded rocks with blocks are in place.. Mottles are small (<3mm) and non-magnetic.

[65]∼20 cm Graded spotted anorthosite with irregular contacts and variably recrystallized pyroxenite blocks/blebs. Proper mottled anorthosite in float; non-magnetic. In field interpreted as Upper/main mottled anorthosite (and related package).

[66]Spotted Anorthosite. ∼8cm thick, with prominent igneous layering and including small amounts of pyroxenite below. Interpreted as UMA. I would not call this a mottled anorthosite, the pyroxenes are do not immediately appear oikocrystic.

Table A.1 (continued)

Name	Ht.	Lat.	Lon.	Elev.	Unit	Lith.	Structure	Photos
XENOLITH[67]	1035	-25.17860	30.07051	1946	Mz	Sa	–	91
UMA2	1035	-25.17640	30.07222	1967	Mz	Sa	–	–
D8ND[68]	1036	-25.17900	30.07014	1936	Mz	Lg	M.L. 110/10	85
D8ND[69]	1036	-25.17900	30.07014	1936	Mz	Hg	Clv. 210/10	–
D8NA[70]	1244	-25.17050	30.06651	1814	Mz	Hg	–	84
D6SE[71]	1417	-25.21910	30.02170	1771	Mz	A	–	–
D6SF[72]	1417	-25.21910	30.02170	1771	IS	IS	–	59
D10NC	1468	-25.20220	30.03204	1755	Mz	Hg	Clv. 180/10	–
MZ31	1473	-25.20990	30.02600	1756	Mz	Hg	–	–
D10NB[73]	1475	-25.20090	30.03275	1781	Mz	Hg	Clv. 200/10	–
D6NB[74]	1551	-25.21840	30.01618	1816	Mz	Hg	Clv. -74/6	58
D10ND	1561	-25.19970	30.02966	1772	Mz	Hg	Clv. 225/15	–
MZD1	1562	-25.21950	30.01472	1780	Mz	Hg	–	–
D10NA	1582	-25.19910	30.02914	1767	Mz	Hg	Clv. 200/10	–
MZD2	1623	-25.22540	30.00751	1809	Mz	Hg	–	–
MZ4	1778	-25.19310	30.02448	1733	Mz	Hg	–	–
D10NH1[75]	2050	-25.18090	30.02113	1746	IS	IS	–	–
D10NE	2056	-25.17710	30.02374	1757	Mz	Hg	Clv. 130/10	–
MZ12	2057	-25.17750	30.02334	1740	Mz	Hg	–	–
D10NH[76]	2136	-25.17770	30.01951	1729	IS	IS	–	–
MZ21	2141	-25.17190	30.02370	1746	Mz	Hg	–	–
D10NG[77]	2175	-25.17710	30.01813	1726	Mz	A	–	–
D10NF	2182	-25.17820	30.01698	1730	Mz	Hg	Clv. 225/15	–
D6ND[78]	2682	-25.21330	29.96669	1526	Mz	Hg	Clv. 160/13	–
D6NC[79]	3266	-25.19900	29.95029	1476	Mz	Hg	Clv. 202/12	–

A.2 Annotated Field Photographs

The annotated field-photograph album is omitted from this abbreviated PDF to keep the file small; it is included in the full dissertation.

[67] A second xenolith near photo 91.

[68] Layered Gabbronorite. Centimeter-scale igneous layering with variable color and grain size.

[69] Homogeneous gabbronorite. Different bedding in nearby unlayered rocks...

[70] Porphyritic gabbronorite. Blebby Xenolith noted and photographed.

[71] Green gabbronorite with amphibole "mottles". Amphibole mottles (<5 cm) are strongly imbricated, undulose shine, not perfectly optically continuous.

[72] Ironstone concretion. <2 cm Goethite nodules enclosing limonite cores with Hematite-Limonite matrix. Rounded clastic quarts <2 mm, in float.

[73] Included a range of measured strikes from 250-315, on fault?

[74] Measured another dip (slump?) indirectly from road: 050:30.

[75] Ironstone Concretion in place.

[76] Ironstone Concretion in float.

[77] Amphibole 'mottled' gabbronorite in float. Fault?

[78] Dark and Light Facies.

[79] Dark Facies.

Appendix B

THE BUSHVELD COMPLEX: THE STATE OF KNOWLEDGE AND AREAS OF ONGOING DEBATE

The Paleoproterozoic Rustenburg Layered Series (RLS) of the Bushveld Magmatic Province in northeast South Africa is the largest layered igneous intrusion in the world, and contains the world's largest source of platinum group elements as well as important reserves of chromium, vanadium, titanium, and iron. It is exceptionally well-preserved, and has been the subject of intense study and debate for generations. Studies focusing on the intrusion's margin have been especially fruitful in informing geologists' understanding of igneous petrogenesis and intrusion mechanisms, modes, and tempos of layered mafic intrusions and plutonic rocks more generally.

B.1 Layered Mafic Intrusions

Layered mafic intrusions (LMIs) are compositionally zoned formations of mafic rocks characterized by distinct millimeter- to meter-scale layering of igneous minerals. Traditionally, the layering and chemical zonation of these geologic marvels have been attributed to processes such as fractional crystallization and gravitational settling within a magma chamber (Wager et al., 1960; Charlier et al., 2015). Notable examples of LMIs include the Stillwater (2.71 Ga) in Montana, USA; Great Dyke (2.57 Ga) in central Zimbabwe; the Bushveld's layered series (2.06 Ga) in South Africa's Limpopo, Mpumalanga, and Gauteng provinces; the Muskox intrusion (1.27 Ga) in Nunavut, Canada; the Duluth Complex (1.01 Ga) in Minnesota, USA; the Sept Iles intrusion (565 Ma) in Quebec, Canada; Dufek (189 Ma) in Antarctica; the Rum intrusion (60.5 Ma) in the Inner Hebrides, Scotland; and the Skaergaard intrusion (55.4 Ma) in East Greenland. Despite their significance as supposedly simple natural laboratories for studying the chemical evolution of magmas, numerous aspects of their formation remain subject to debate, even including the presence of significant liquid magma in their formation (W. Kruger and Latypov, 2020; Mungall et al., 2016; Latypov et al., 2022; Yao et al., 2021).

Experimental observations have revealed that plagioclase flotation occurs in mafic magmas at depths corresponding to the Earth's crust (Campbell et

al., 1978). This challenges the notion that gravitational settling alone is responsible for igneous layering. Instead, chemical means of phase separation are often suggested, referencing processes like periodic nucleation during in-situ crystallization (McBirney and Noyes, 1979), in-situ crystallization of a magma with compositional layering developed and sustained through double-diffusive convection (Huppert and Turner, 1981), or preferential dissolution and recrystallization of existing unstructured crystal cumulates or mushes to minimize crystal surface energy perpendicular to temperature gradients (A. Boudreau, 2011) or by preferential attack by chlorine-rich fluids (Sonnenthal, 1992). An alternative mechanical explanation, seismically aided hydrodynamic sorting within a tilting, crystal-rich cumulate pile, has also enjoyed support (Maier et al., 2013).

B.2 The Bushveld Event and RLS

The Rustenburg Layered Series (RLS) is renowned as the world's largest layered mafic intrusion, distinguished by its remarkably continuous layers, sometimes across hundreds of kilometers (Cawthorn, 2015; Latypov et al., 2022; Scoon and Teigler, 1994). Within the RLS, pegmatoid "reefs" and chromitites host the world's largest reserves of platinum group elements or PGEs (Lee, 1996). In addition to the RLS, the Bushveld event encompasses several satellite intrusions, including the Uitkomst Intrusion, Phalaborwa Carbonatite, the Stavoren Granophyre at the top of the RLS, the massive felsic Rooiberg lava eruption covering 200,000 km^3, and the world's largest A-type granitic pluton at the intrusion's roof: the Lebowa granite suite (Hill et al., 1996; Kleemann and Twist, 1989; Scoates et al., 2021; Walraven, 1985).

The mafic plutonic rocks of the RLS are subdivided into the Lower Zone, Critical Zone, Main Zone, and Upper Zone, with unlayered sills and chill sequences at the margin known as the Marginal Zone (von Gruenewaldt et al., 1985). These zones are primarily distinguished by their mineralogy, which transitions from more primitive assemblages in the Lower Zone to more evolved ones in the Upper Zone. The Lower Zone uniquely features dunites and harzburgites, the Critical Zone features chromitites and norites, the Main Zone features gabbronorites, and lastly the Upper Zone features diorites and magnetitites. The RLS exhibits significant compositional zoning, with increasing iron, silica, vanadium, and alkali content as well as chromium and magnesium depletion toward the intrusion's roof (Cawthorn, 2015). While the RLS often maintains remarkable

lateral continuity on the kilometer scale, variations in stratigraphy, PGE mineralization, and thickness sometimes present challenges to stratigraphic correlations (Ashwal et al., 2005; McDonald et al., 2005; Scoon and Teigler, 1994).

B.3 Structure and Relationship to Country Rocks

The layered series outcrops as distinct lobes on the present-day surface, with variations in stratigraphy, PGE mineralization, and mineralogy observed across limbs. These lobes are separated and influenced by geological features like the Thabazimbi-Murchison and Steelpoort lineaments (du Plessis and Walraven, 1990; Ashwal et al., 2005; McDonald et al., 2005; Scoon and Teigler, 1994). Despite these variations, there is remarkable continuity between layers. Chromitite layers with similar thickness and chemical signatures can be correlated across limbs separated by hundreds of kilometers of felsic igneous rocks and sedimentary cover. Additionally, crustal thickness models derived from seismic receiver functions and gravity anomalies support the possibility that the intrusion is connected at depth (Webb et al., 2004; Cawthorn and Webb, 2001). Some researchers have further classified the intrusion into “distal” and “proximal” facies based on the hypothesis that the intrusion's feeders exploited the Thabazimbi-Murchison Lineament, supported by observations such as the best development of ultramafic zones closest to this lineament (Scoon and Teigler, 1994; B. M. Clarke et al., 2000).

The RLS generally dips inward toward its center, with the most significant dips occurring at the margin (Button, 1976; Frick, 1967). While the RLS was historically considered a lopolith (H. D. B. Wilson, 1956), paleomagnetic studies favor an initially horizontal orientation that flexed isostatically after crystallization ceased (Hattingh, 1986). The floor of the RLS exhibits synmagmatic deformation, possibly resulting from interactions during the coalescence of offset interfingered sills (Pollard et al., 1975; B. Clarke et al., 2009). Abundant large sedimentary inliers within the complex can also be explained by the inhomogeneous propagation of finger sills along the complex's margin (B. M. Clarke et al., 2000; B. Clarke et al., 2009).

B.4 Isotopic and Volatile Constraints on the RLS

Radiogenic Isotope Stratigraphy of the RLS

F. J. Kruger (1994) reported a comprehensive strontium isotope stratigraphy of the RLS, revealing radiogenic Sr isotopes indicative of varying degrees of crustal contamination throughout the intrusion. This stratigraphy differentiates the Lower, Critical, and lower Main Zones from the upper Main Zone and Upper Zone. The lower portion of the intrusion records highly variable Sr isotopes, interpreted to arise due to repeated magma replenishments in an open system. F. J. Kruger (1994) refers to this zone as the "Integration Stage." Conversely, the middle Main Zone displays fairly consistent isotopic compositions, transitioning to more primitive values at the Pyroxenite Marker—a well-known bed in the upper Main Zone signifying a shift towards more primitive magmas (Cawthorn et al., 1991). These two stratigraphic regions with internally constant initial strontium isotope ratios are referred to as the "Differentiation Stage" and could possibly have evolved through closed-system fractional crystallization on the basis of strontium isotopes. Studies on neodymium isotopes within the RLS are less detailed than those available for strontium isotopes, but show a broadly similar pattern that requires crustal contamination throughout the entire complex, with the Main and Upper Zones requiring more radiogenic (crustal?) contamination than the Lower and Critical Zones (Maier et al., 2000).

Strontium isotope disequilibrium is well-known within the Merensky and Bastard reefs, where pyroxenes and plagioclase have different and variable initial strontium isotope values (Seabrook et al., 2005; Cawthorn, 2015); however, strontium isotope disequilibrium is not unique to the Merensky and Bastard reefs. Chutas et al. (2012) demonstrated that plagioclase throughout the Lower and Critical Zones was isotopically zoned with primitive cores and evolved rims, and that coexisting pyroxenes and plagioclase were not in isotopic equilibrium. Roelofse and Ashwal (2012) similarly observed significant strontium isotopic disequilibrium in lower Main Zone rocks. This means that crystalline phases from the entire Lower, Critical, and lower Main Zone cumulates interacted with exotic magmas or hydrothermal fluids after substantial crystallization, either by later intrusion of an exotic magma into a preexisting crystal mush at the level of the RLS (Chutas et al., 2012) or by intrusion of xenocryst-bearing crystal-rich magmas derived from a subcompartmentalized chamber at depth (Roelofse and Ashwal, 2012). Any rock in the Critical, Lower, and lower Main Zones

was thus formed by multiple magmatic ± hydrothermal events/environments, and any bulk isotopic or chemical measurement of these zones will inherently integrate over multiple crystallization episodes.

A comprehensive dataset of rhenium-osmium (Re-Os) isotopes provides valuable insights into the history of platinum group elements (PGEs) within the Bushveld Complex. Schoenberg et al. (1999) measured a 2.043 ± 0.011 Ga isochron age for the Bastard Reef, which contained little common osmium. Furthermore, they report a systematic trend in the initial $^{187}Os/^{188}Os$ ratios within chromitites spanning the Critical Zone. These ratios exhibit a distinct progression, starting from values closely resembling chondritic compositions and escalating towards highly radiogenic values as one moves upwards through the stratigraphic sequence reaching their most radiogenic values at the Merensky Reef and upper group (UG) chromitites (Schoenberg et al., 1999; McCandless et al., 1999).

The presence of heterogeneous initial $^{187}Os/^{188}Os$ values within chromitites is commonly invoked to support the argument for an orthomagmatic origin for PGE mineralization in the upper Critical Zone, arguing against a hydrothermal origin (Schoenberg et al., 1999; McCandless et al., 1999). However, the validity of the argument depends upon whether Re-Os isotopes faithfully record magmatic processes. Notably, experimental evidence has demonstrated that osmium diffuses rapidly within sulfides, displaying a closure temperature similar to the closure temperature of argon within biotite (Brenan et al., 2000). Ar-Ar thermochronology of biotite therefore can be used in this case as a proxy to determine the timing of the closure of the Re-Os system. Nomade et al. (2004) reported an average Ar-Ar age of 2.042 ± 0.003 Ga for biotite within the UG2 chromite, which agrees well with the Re-Os isochron age of the Bastard Reef. Furthermore, this congruence extends beyond the Bushveld Complex, as similar Ar-Ar ages are observed on a regional scale within syn-kinematic micas found within low-grade phyllitic metasediments of the Transvaal Supergroup (Alexandre et al., 2006). These Ar-Ar ages effectively date regional compression within the Kaapvaal craton, including deformation into fold and thrust belts (Alexandre et al., 2006). Consequently, it can be inferred that the Re-Os isotope system underwent diffusive reequilibration within the chromitites approximately 10 million years after the cessation of magmatism within the RLS, which ended at 2.055 Ga (age derived from Scoates et al., 2021; Zeh et al., 2015).

The most enigmatic aspect of the Re-Os isotope geochemistry of the RLS

centers around the coexistence of two unusual, isotopically distinct sulfide populations within the Merensky Reef. A substantial population of laurite (RuS_2) exhibits the most radiogenic initial $^{187}Os/^{188}Os$ values of the entire 8-kilometer stratigraphy of the RLS, while the same hand sample contains erlichmanite (OsS_2) with an osmium isotopic ratio mirroring a chondritic value, the least radiogenic $^{187}Os/^{188}Os$ within the entire RLS (Hart and Kinloch, 1989). It is difficult to imagine how sulfides with such different $^{187}Os/^{188}Os$ values were able to coexist within the same rock without isotopically equilibrating.

In a broader geological context, Richardson and Shirey (2008) conducted measurements of sulfide Re-Os isotopes, as well as silicate Rb-Sr and Sm-Nd isotopes, within kimberlitic diamond inclusions in close proximity to the Bushveld Complex. Their aim was to constrain the isotopic reservoirs within the subcontinental lithospheric mantle (SCLM) beneath the Bushveld Complex. Their findings reveal the enrichment of radiogenic isotopes across all three isotopic systems within the SCLM, which they use to argue that the radiogenic isotopic signatures observed within the RLS can be explained as a three-part mixture of mantle reservoirs that requires no contributions from the crust, instead requiring only the harzburgitic SCLM, eclogitic SCLM, and the depleted, convecting mantle.

Stable Isotopes of the RLS

Oxygen isotopic studies show that oxygen within the RLS was equilibrated at magmatic temperatures, remarkably unaffected by interactions with surface-derived fluids (Schiffries and Rye, 1989; Harris et al., 2005). Early studies indicated magmatic temperatures for the Upper Zone equilibrating at 1000°C, with the rest of the intrusion at 1200°C (Schiffries and Rye, 1989). However, later studies produced scattered equilibration temperatures as low as 550°C (Harris et al., 2005). Oxygen isotopes within the RLS exhibit remarkable consistency throughout the intrusion and are slightly heavier (~8‰ $\delta^{18}O$) than mantle values (~6‰ $\delta^{18}O$) suggesting substantial but well-mixed contamination from surface-derived material, possibly from Transvaal Supergroup rocks exposed on the surface or from unexposed, isotopically heavy rocks at depth (Schiffries and Rye, 1989; Harris et al., 2005). Similarly, multiple-sulfur isotopes identified a well-mixed and uniform non-zero mass independent fractionation (MIF) of $\Delta^{33}S$ (Penniston-Dorland et al., 2012) across the RLS, requiring a component of surface-derived sulfur that was very efficiently mixed

throughout the entirety of the RLS. The remarkable homogeneity in volatile element isotopes throughout the RLS contrasts with heterogeneous strontium and neodymium isotopes throughout the intrusion, alluding to a decoupling of magmatic and hydrothermal material sources within the RLS. The volatile element isotopic data require a single source for late-stage hydrothermal fluids across the complex.

Hydrogen isotopes indicate late-stage hydrothermal metasomatism at low water/rock ratios (Schiffries and Rye, 1990). Carbonated fluid within the RLS imparted a carbon isotope signature likely arising from the devolatilization of Transvaal carbonates (Schiffries and Rye, 1990). Sedimentary country rocks display oxygen and carbon isotopic depletions correlating with increasing metamorphic grade and proximity to the RLS margins (Buick et al., 2000; Schiffries and Rye, 1990). The lack of complete isotopic equilibration of metasedimentary country rocks and RLS fluids suggests that fluids flowed from devolatilized metasedimentary rocks into the RLS, and not vice versa (Buick et al., 2000).

Unusual Volatile-bearing Phases in the Lower RLS

Within the Lower and Critical Zones of the RLS, volatile-bearing phases such as apatite, amphibole, and phlogopite are rare but often contain exceptionally high chlorine concentrations (A. E. Boudreau et al., 1986; Willmore et al., 2000). These phases are most concentrated in the lateral Merensky Reef and the vertical metasomatic Driekop dunite pipe, often associated with magmatic graphite and platiniferous sulfides (Ballhaus and Stumpfl, 1985; Schiffries, 1982). Thermodynamic calculations suggest that the initial volatile phase exsolved from RLS magmas was predominantly hydrochloric acid (HCl) and chloromethane (CH_3Cl), with minor amounts of water and sulfur (Mathez et al., 1989). This fluid likely exsolved sulfur during graphite crystallization. Chlorine isotopic studies of the RLS show that these Cl-rich fluids were not derived from the local Transvaal Supergroup, but are consistent with an origin from SCLM charged with volatiles from dehydrating subducting altered oceanic crust (Willmore et al., 2002). The association of these C-Cl rich volatile phases with platiniferous horizons has led many authors to suggest that PGE mineralization is controlled by C-Cl rich fluids, perhaps by chromatographic extraction by reactive porous flow of C-Cl rich fluids within the existing Critical Zone cumulates (A. E. Boudreau and Meurer, 1999).

B.5 Petrological Debates Within the RLS

Open or Closed System

Another quintessential LMI, the Skaergaard intrusion, is thought to have formed primarily through closed-system fractional crystallization from the chamber's outer regions inward (McBirney and Noyes, 1979), albeit with strontium isotopic evidence of country rock assimilation in its early stages (Stewart and DePaolo, 1990). In contrast, the RLS is interpreted as having crystallized mainly in an open system, featuring frequent magmatic recharge. This is supported by numerous chemical reversals in plagioclase and pyroxene compositions, as well as variations in initial strontium isotopes within the Lower, Critical, and lower Main Zones (F. J. Kruger, 1994; Cawthorn, 2015). However, the upper Main and Upper Zones (sometimes lumped together as the UUMZ) of the RLS are often seen as products of closed-system fractional crystallization, as indicated by uniform strontium isotopes (F. J. Kruger, 1994), oxygen isotopes that seem to have evolved through closed-system fractional crystallization (Bucholz et al., 2017), and well-fitting crystallization sequences produced by MELTS simulations of closed-system fractional crystallization of plausible UUMZ parental compositions (VanTongeren et al., 2010). Nevertheless, this view has been challenged by the identification of significant reversals in plagioclase and non-conformable field relationships within the UUMZ (Yuan et al., 2017; Scoon and Mitchell, 2012).

Spatial and Temporal Scales of Magma Chambers in the Crust, Pluton Assembly Mechanisms

The RLS, along with other LMIs, has become a battleground for debates regarding the temporal and spatial scales of crustal magma chambers and pluton assembly. Some argue that plutonic rocks represent fossilized magma chambers (W. Kruger and Latypov, 2020; Bachmann and Bergantz, 2004; Paterson, 2009), while others propose that these rocks are assembled incrementally through sill-like intrusions. Such arguments are applied to rocks within the RLS by Mitchell et al. (2019), Yao et al. (2021), Scoon and Mitchell (2023), and Bourdeau et al. (2023). The latter viewpoint first gained significant support when detailed, high-precision zircon U/Pb geochronology studies revealed wide ranges of crystallization ages within seemingly homogeneous, continuous plutons (Coleman et al., 2004; Glazner et al., 2004). These wide age ranges produced by zircon U/Pb geochronology in plutons greatly exceed the timeframes over

which large magma chambers can be thermally sustained before cooling (Annen, 2009; Cawthorn and Walraven, 1998).

Zircons from the RLS and Stillwater layered mafic intrusions similarly yield some out-of-sequence U/Pb ages that challenge the idea that they represent large, fossilized magma chambers (Mungall et al., 2016; Wall et al., 2018; Scoates et al., 2021). However, resolvably out-of-sequence zircons are somewhat rare and were not detected in a slightly earlier study (Zeh et al., 2015). Additionally, some authors suggest these data are impossible to reconcile with some field relationships, challenging the reliability of U/Pb zircon geochronology for investigating layered mafic intrusions (Latypov and Chistyakova, 2022). Nevertheless, U/Pb zircon geochronology successfully yields narrow age ranges for other igneous rocks when these are geologically expected, such as for erupted ignimbrites (Wotzlaw et al., 2013).

Origin of Enigmatic Monomineralic Igneous Rocks

The RLS contains enigmatic monomineralic igneous rocks that continue to perplex igneous petrologists. For instance, the RLS contains dozens of centimeter to meter scale, near-pure chromitites which are challenging to explain. One common explanation involves the mixing of primitive and evolved magmas inducing chromite saturation (Irvine, 1977). However, an in-situ crystallization hypothesis is problematic because the Lower and Critical Zones contain significantly more chromium than could have been delivered by a plausible amount of chromium-saturated silicate melt (Eales, 2000), and the cumulate rocks above and below some chromitites are petrologically identical (Mondal and Mathez, 2007), seeming to preclude in-situ magma mixing unless over 300,000 km^3 of differentiated mafic lavas were lost from the chamber's top (Cawthorn and Walraven, 1998). The most popular alternative explanation posits that chromite crystallized within staging chambers and was injected into the RLS as chromite-phyric silicate magmas (Mondal and Mathez, 2007; Eales, 2000). Experiments have shown that chromitites can form through magmatic sedimentation from a chromite-bearing magma within realistic timescales (Manoochehri and Schmidt, 2014).

Stratiform anorthosites within the RLS pose another challenge to conventional explanations because plausible RLS magma compositions are rarely, if ever, saturated in plagioclase alone (Latypov et al., 2020; Maier et al., 2020). Various

hypotheses propose that these anorthosites resulted from [1] density-driven mechanical sorting of a magma concomitantly saturated in multiple phases (Cawthorn and Ashwal, 2009); [2] depressurization-forced saturation of plagioclase alone, followed by crystal settling (Yuan et al., 2017) or in-situ growth on the magma chamber floor (Latypov et al., 2020); [3] tectonically-induced mechanical sorting of a preexisting cumulate pile (Maier et al., 2016); [4] reactive porous flow of acidic fluids (Maier et al., 2020; Sonnenthal, 1992); or [5] restites from partial melting at the contact of new intrusions of magma (Mitchell et al., 2019).

The Origin of the Merensky Reef and PGE Mineralization

The origin of the Merensky Reef and PGE mineralization in the upper Critical Zone remains contentious. One theory suggests that immiscible sulfide fluids scavenged PGEs from a large magma chamber (Campbell et al., 1983). Alternatively, percolating fluids rich in chlorine, carbon, and sulfur may have redistributed PGE elements within an already-solidified Critical Zone (A. E. Boudreau et al., 1986; Ballhaus and Stumpfl, 1985), or magmatic differentiation controlled by chromite may have concentrated PGEs within this zone (Scoon and Teigler, 1994). The platiniferous Merensky Reef, a two-part pyroxenite and pegmatoidal horizon, is debated to be either a composite of multiple synmagmatic or out-of-sequence sills (Mitchell and Scoon, 2007; Mitchell et al., 2019) or a mixture of disequilibrated crystals precipitating synchronously from overlying density-stratified Main and Critical Zone magmas (Seabrook et al., 2005). Most authors agree that the Merensky Reef represents a mixture of Main Zone and Critical Zone magmas (F. J. Kruger, 1994; Sharpe, 1985; Cawthorn, 2015).

The Transformation Hypothesis

van Biljon (1949) developed a controversial hypothesis for the origin of the RLS: that it represented an extreme case of in-situ metasomatism of the upper Pretoria Group sedimentary rocks of the Transvaal Supergroup on the basis of field relationships and geochemical arguments. Indeed, terminology and concepts from sedimentology and stratigraphy are often (though arguably erroneously) employed to describe features within the RLS and other LMIs. Van Biljon's Transformation Hypothesis sought to explain the origin of the enormously laterally continuous markers within the RLS, the apparent problem

of an enormous amount of displaced sedimentary rocks, and the transgressive nature of the RLS's contact with the Pretoria Group. van Biljon (1949) argues that chert is transformed to anorthosites, carbonates are transformed to magnetitites/chromitites, variably impure quartzites are transformed to (gabbro)norites, and shales are transformed to pyroxenites. He presents geochemical evidence that seems to show a chemical continuum in different oxide spaces between unmetamorphosed sedimentary rocks, variably hornfelsized sedimentary rocks, and mafic rocks of the RLS.

Button (1976) emphasized Van Biljon's recognition of the "space problem" introduced by the transgressive RLS into relatively undeformed sedimentary rocks, instead arguing that the felsic Bushveld rocks represented the displaced (and transformed) sediments. However, Cawthorn (1998) showed through a simple geometric analysis that the apparently missing sedimentary rocks may have been preserved underneath the RLS given the nature of its transgressive and discordant contact with the Pretoria Group.

One simple counterpoint to Van Biljon's transformation hypothesis for the RLS is that similar LMIs exist that are emplaced entirely or nearly entirely within non-sedimentary granitic, basaltic, and gneissic country rock, such as the chromitite-bearing Great Dyke of Zimbabwe (Worst, 1958) and the finely-layered Skaergaard Intrusion (McBirney, 1996). Furthermore, the RLS itself onlaps granite in the North Limb (van der Merwe, 1976).

Generally, available stable and radiogenic isotopic data require whole-intrusion-scale magmatic and hydrothermal processes to be the dominant processes forming the RLS, and for local contamination with crustal rocks to play a subordinate role (Zirakparvar et al., 2014; Schiffries and Rye, 1989; Schiffries and Rye, 1990; Willmore et al., 2002; Penniston-Dorland et al., 2012; F. J. Kruger, 1994; Richardson and Shirey, 2008). This general statement has notable exceptions:

1. The Marginal Zone contains abundant metapelitic migmatites and calc-silicates (Johnson et al., 2003; van Biljon, 1949; Button, 1976; Frick, 1967, and Chapter 5 of this work).

2. Interstitial phases such as anhydrite, almost certainly derived from sedimentary rocks, are documented within the ultramafic Lower Zone (A. H. Wilson et al., 2017; Yudovskaya et al., 2021).

These observations together suggest that in-situ igneous sediment transformation is not an essential or dominant process in the formation of the RLS or other LMIs, but that it may still occur at small scales. Evidence for this process would become obscured and eventually destroyed by episodic melting and/or reaction with an abundance of incoming magmas and hydrothermal fluids, suggesting that this process could also have been locally important at higher levels within the RLS even when evidence is subtle or completely lost to igneous overprinting.

REFERENCES

Alexandre, P., M. A. G. Andreoli, A. Jamison, and R. L. Gibson (2006). "$^{40}Ar/^{39}Ar$ age constraints on low-grade metamorphism and cleavage development in the Transvaal Supergroup (central Kaapvaal craton, South Africa): implications for the tectonic setting of the Bushveld Igneous Complex". In: *South African Journal of Geology* 109, pp. 393–410.

Allmendinger, R. W., N. Cardozo, and D. M. Fisher (2012). *Structural Geology Algorithms: Vectors and Tensors*. Cambridge: Cambridge University Press.

Annen, C. (2009). "From plutons to magma chambers: Thermal constraints on the accumulation of eruptible silicic magma in the upper crust". In: *Earth and Planetary Science Letters* 284, pp. 409–416.

Ashwal, L. D., S. J. Webb, and M. W. Knoper (2005). "Magmatic stratigraphy in the Bushveld Northern Lobe: continuous geophysical and mineralogical data from the 2950 m Bellevue drillcore". In: *South African Journal of Geology* 108, pp. 199–232.

Bachmann, O. and G. W. Bergantz (2004). "On the Origin of Crystal-poor Rhyolites: Extracted from Batholithic Crystal Mushes". In: *Journal of Petrology* 45, pp. 1565–1582.

Baker, S. R. and A. E. Boudreau (2019). "The influence of the thick banded series anorthosites on the crystallization of the surrounding rock of the Stillwater Complex, Montana". In: *Contributions to Mineralogy and Petrology* 174, p. 99.

Ballhaus, C. G. and E. F. Stumpfl (1985). "Occurrence and petrological significance of graphite in the Upper Critical Zone, western Bushveld Complex, South Africa". In: *Earth and Planetary Science Letters* 74, pp. 58–68.

Barnes, S. J., R. Latypov, S. Chistyakova, B. Godel, and L. E. Schoneveld (2021). "Idiomorphic oikocrysts of clinopyroxene produced by a peritectic reaction within a solidification front of the Bushveld Complex". In: *Contributions to Mineralogy and Petrology* 176.1, p. 5. DOI: 10.1007/s00410-020-01747-4.

Boudreau, A. (2011). "The evolution of texture and layering in layered intrusions". In: *International Geology Review* 53, pp. 330–353.

Boudreau, A. E., E. A. Mathez, and I. S. McCallum (1986). "Halogen Geochemistry of the Stillwater and Bushveld Complexes: Evidence for Transport of the Platinum-Group Elements by Cl-Rich Fluids". In: *Journal of Petrology* 27, pp. 967–986.

Boudreau, A. E. and W. P. Meurer (1999). "Chromatographic separation of the platinum-group elements, gold, base metals and sulfur during degassing of a compacting and solidifying igneous crystal pile". In: *Contributions to Mineralogy and Petrology* 134, pp. 174–185.

Bourdeau, J. E., B. Hayes, S. E. Zhang, A. Logue, and G. M. Bybee (2021). "Origin and significance of noritic blocks in layered anorthosites in the Bushveld Complex, South Africa". In: *Contributions to Mineralogy and Petrology* 177.1, p. 11. DOI: 10.1007/s00410-021-01872-8.

Bourdeau, J. E., S. E. Zhang, B. Hayes, A. Logue, and G. M. Bybee (2023). "Evidence for the incremental assembly of the Pyroxenite Marker, Bushveld Complex, by the emplacement of crystal slurries". In: *Lithos* 438–439, p. 107007. DOI: 10.1016/j.lithos.2022.107007.

Brenan, J. M., D. J. Cherniak, and L. A. Rose (2000). "Diffusion of osmium in pyrrhotite and pyrite: implications for closure of the Re–Os isotopic system". In: *Earth and Planetary Science Letters* 180, pp. 399–413.

Bucholz, C. E., O. Jagoutz, J. A. VanTongeren, J. Setera, and Z. Wang (2017). "Oxygen isotope trajectories of crystallizing melts: Insights from modeling and the plutonic record". In: *Geochimica et Cosmochimica Acta* 207, pp. 154–184.

Buick, I. S. et al. (2000). "Fluid flow in metacarbonates associated with emplacement of the Bushveld Complex, South Africa". In: *Journal of Geochemical Exploration* 69–70, pp. 391–395.

Button, A. (1976). "Stratigraphy and Relations of the Bushveld Floor in the Eastern Transvaal". In: *Transactions of the Geological Society of South Africa* 79, pp. 3–12.

Cameron, E. N. (1982). "The upper critical zone of the eastern Bushveld Complex; precursor of the Merensky Reef". In: *Economic Geology* 77, pp. 1307–1327.

Campbell, I. H., A. J. Naldrett, and S. J. Barnes (1983). "A Model for the Origin of the Platinum-Rich Sulfide Horizons in the Bushveld and Stillwater Complexes". In: *Journal of Petrology* 24, pp. 133–165.

Campbell, I. H., P. L. Roeder, and J. M. Dixon (1978). "Plagioclase buoyancy in basaltic liquids as determined with a centrifuge furnace". In: *Contributions to Mineralogy and Petrology* 67, pp. 369–377.

Cardozo, N. and R. W. Allmendinger (2013). "Spherical Projections with OSXStereonet". In: *Computers & Geosciences* 51, pp. 193–205.

Cawthorn, R. G. (1998). "Geometrical relations between the Transvaal Supergroup, the Rooiberg Group, and the mafic rocks of the Bushveld Complex". In: *South African Journal of Geology* 101, pp. 275–279.

Cawthorn, R. G. (2015). "The Bushveld Complex, South Africa". In: *Layered Intrusions*. Ed. by B. Charlier, O. Namur, R. Latypov, and C. Tegner. Dordrecht: Springer Netherlands, pp. 517–587. DOI: 10.1007/978-94-017-9652-1_12.

Cawthorn, R. G. and L. D. Ashwal (2009). "Origin of Anorthosite and Magnetitite Layers in the Bushveld Complex, Constrained by Major Element Compositions of Plagioclase". In: *Journal of Petrology* 50, pp. 1607–1637.

Cawthorn, R. G., P. S. Meyer, and F. J. Kruger (1991). "Major Addition of Magma at the Pyroxenite Marker in the Western Bushveld Complex, South Africa". In: *Journal of Petrology* 32, pp. 739–763.

Cawthorn, R. G. and F. Walraven (1998). "Emplacement and Crystallization Time for the Bushveld Complex". In: *Journal of Petrology* 39, pp. 1669–1687.

Cawthorn, R. G. and S. J. Webb (2001). "Connectivity between the western and eastern limbs of the Bushveld Complex". In: *Tectonophysics* 330, pp. 195–209.

Charlier, B., O. Namur, R. Latypov, and C. Tegner, eds. (2015). *Layered Intrusions*. Dordrecht: Springer Netherlands. DOI: 10.1007/978-94-017-9652-1.

Chutas, N. I., E. Bates, S. A. Prevec, D. S. Coleman, and A. E. Boudreau (2012). "Sr and Pb isotopic disequilibrium between coexisting plagioclase and orthopyroxene in the Bushveld Complex, South Africa: microdrilling and progressive leaching evidence for sub-liquidus contamination within a crystal mush". In: *Contributions to Mineralogy and Petrology* 163, pp. 653–668.

Clarke, B., R. Uken, and J. Reinhardt (2009). "Structural and compositional constraints on the emplacement of the Bushveld Complex, South Africa". In: *Lithos* 111, pp. 21–36.

Clarke, B. M., R. Uken, and M. K. Watkeys (2000). "Intrusion mechanisms of the southwestern Rustenburg Layered Suite as deduced from the Spruitfontein inlier". In: *South African Journal of Geology* 103, pp. 120–127.

Cole, J., C. A. Finn, and S. J. Webb (2013). "Overview of the magnetic signatures of the Palaeoproterozoic Rustenburg Layered Suite, Bushveld Complex, South Africa". In: *Precambrian Research* 236, pp. 193–213. DOI: 10.1016/j.precamres.2013.07.017.

Coleman, D. S., W. Gray, and A. F. Glazner (2004). "Rethinking the emplacement and evolution of zoned plutons: Geochronologic evidence for incremental assembly of the Tuolumne Intrusive Suite, California". In: *Geology* 32, pp. 433–436.

Cousins, C. A. (1959). "The Structure of the Mafic Portion of the Bushveld Igneous Complex". In: *Transactions of the Geological Society of South Africa* 62, pp. 179–189.

du Plessis, C. P. and F. Walraven (1990). "The tectonic setting of the Bushveld Complex in Southern Africa, Part 1. Structural deformation and distribution". In: *Tectonophysics* 179, pp. 305–319.

Eales, H. V. (2000). "Implications of the chromium budget of the Western Limb of the Bushveld Complex". In: *South African Journal of Geology* 103, pp. 141–150.

Eales, H. V., J. S. Marsh, A. A. Mitchell, W. J. de Klerk, F. J. Kruger, and M. Field (1986). "Some Geochemical Constraints upon Models for the Crystallization of the Upper Critical Zone–Main Zone Interval, Northwestern Bushveld Complex". In: *Mineralogical Magazine* 50, pp. 567–582.

Frankel, J. J. (1942). "Chrome-Bearing Magnetic Rocks from the Eastern Bushveld". In: *South African Journal of Science* 38, pp. 152–157.

Frick, C. (1967). "The Margin of the Bushveld Complex in the Vicinity of De Berg, North of Dullstroom". M.Sc. thesis. University of Pretoria.

Glazner, A. F., J. M. Bartley, D. S. Coleman, W. Gray, and R. Z. Taylor (2004). "Are Plutons Assembled over Millions of Years by Amalgamation from Small Magma Chambers?" In: *GSA Today* 14.4, pp. 4–11.

Harmer, R. E. and M. R. Sharpe (1985). "Field relations and strontium isotope systematics of the marginal rocks of the eastern Bushveld Complex". In: *Economic Geology* 80, pp. 813–837.

Harris, C., J. J. M. Pronost, L. D. Ashwal, and R. G. Cawthorn (2005). "Oxygen and Hydrogen Isotope Stratigraphy of the Rustenburg Layered Suite, Bushveld Complex: Constraints on Crustal Contamination". In: *Journal of Petrology* 46, pp. 579–601.

Hart, S. R. and E. D. Kinloch (1989). "Osmium isotope systematics in Witwatersrand and Bushveld ore deposits". In: *Economic Geology* 84, pp. 1651–1655.

Hattingh, P. J. (1986). "The palaeomagnetism of the main zone of the eastern Bushveld Complex". In: *Tectonophysics* 124, pp. 271–295.

Hill, M., F. Barker, D. Hunter, and R. Knight (1996). "Geochemical Characteristics and Origin of the Lebowa Granite Suite, Bushveld Complex". In: *International Geology Review* 38, pp. 195–227.

Holness, M. B., R. G. Cawthorn, and J. Roberts (2017). "The thickness of the crystal mush on the floor of the Bushveld magma chamber". In: *Contributions to Mineralogy and Petrology* 172.11, p. 102. DOI: 10.1007/s00410-017-1423-4.

Huppert, H. E. and J. S. Turner (1981). "Double-diffusive convection". In: *Journal of Fluid Mechanics* 106, pp. 299–329.

Irvine, T. N. (1977). "Origin of chromitite layers in the Muskox intrusion and other stratiform intrusions: A new interpretation". In: *Geology* 5, pp. 273–277.

Johnson, T. E., R. L. Gibson, M. Brown, I. S. Buick, and I. Cartwright (2003). "Partial Melting of Metapelitic Rocks Beneath the Bushveld Complex, South Africa". In: *Journal of Petrology* 44, pp. 789–813.

Karykowski, B. T. et al. (2017). "In situ Sr Isotope Compositions of Plagioclase from a Complete Stratigraphic Profile of the Bushveld Complex, South Africa: Evidence for Extensive Magma Mixing and Percolation". In: *Journal of Petrology* 58, pp. 2285–2308.

Khoza, B. (2021). *Tennis Ball Marker and Related PGE Mineralization in the Eastern Bushveld Complex, South Africa*.

Kleemann, G. J. and D. Twist (1989). "The Compositionally-zoned Sheet-like Granite Pluton of the Bushveld Complex: Evidence Bearing on the Nature of A-type Magmatism". In: *Journal of Petrology* 30, pp. 1383–1414.

Klemm, D. D., S. Ketterer, F. Reichhardt, J. Steindl, and K. Weber-Diefenbach (1985). "Implication of vertical and lateral compositional variations across the pyroxene marker and its associated rocks in the upper part of the main zone in the eastern Bushveld Complex". In: *Economic Geology* 80, pp. 1007–1015.

Kruger, F. J. (1994). "The Sr-Isotopic Stratigraphy of the Western Bushveld Complex". In: *South African Journal of Geology* 97, pp. 393–398.

Kruger, W. and R. Latypov (2020). "Fossilized solidification fronts in the Bushveld Complex argue for liquid-dominated magmatic systems". In: *Nature Communications* 11, p. 2909.

Latypov, R. et al. (2020). "Monomineralic anorthosites in layered intrusions are indicators of the magma chamber replenishment by plagioclase-only-saturated melts". In: *Scientific Reports* 10, p. 3839.

Latypov, R. and S. Chistyakova (2022). "Misinterpretation of zircon ages in layered intrusions". In: *South African Journal of Geology* 125, pp. 13–26.

Latypov, R., S. Chistyakova, R. A. Hornsey, G. Costin, and M. van der Merwe (2022). "A 5-km-Thick Reservoir with >380,000 km^3 of Magma within the Ancient Earth's Crust". In: *Scientific Reports* 12, p. 15651.

Lee, C. A. (1996). "A Review of Mineralization in the Bushveld Complex and Some Other Layered Intrusions". In: *Layered Intrusions*. Ed. by R. G. Cawthorn. Vol. 15. Developments in Petrology. Amsterdam: Elsevier, pp. 103–145.

Lee, C. A. and M. R. Sharpe (1979). "Spheroidal pyroxenite aggregates in the Bushveld Complex – a special case of silicate liquid immiscibility". In: *Earth and Planetary Science Letters* 44, pp. 295–310.

Maier, W. D. et al. (2020). "Formation of Bushveld anorthosite by reactive porous flow". In: *Contributions to Mineralogy and Petrology* 176, p. 3.

Maier, W. D., N. T. Arndt, and E. A. Curl (2000). "Progressive crustal contamination of the Bushveld Complex: evidence from Nd isotopic analyses of the cumulate rocks". In: *Contributions to Mineralogy and Petrology* 140, pp. 316–327.

Maier, W. D. and S.-J. Barnes (2024). "Origin of Chromitite-Anorthosite Interlayering in the Bushveld Complex". In: *The Canadian Journal of Mineralogy and Petrology* 62, pp. 731–745.

Maier, W. D., S.-J. Barnes, and D. I. Groves (2013). "The Bushveld Complex, South Africa: formation of platinum–palladium, chrome- and vanadium-rich layers via hydrodynamic sorting of a mobilized cumulate slurry in a large, relatively slowly cooling, subsiding magma chamber". In: *Mineralium Deposita* 48, pp. 1–56.

Maier, W. D., B. T. Karykowski, and S.-H. Yang (2016). "Formation of transgressive anorthosite seams in the Bushveld Complex via tectonically induced mobilisation of plagioclase-rich crystal mushes". In: *Geoscience Frontiers* 7, pp. 875–889.

Manoochehri, S. and M. W. Schmidt (2014). "Settling and compaction of chromite cumulates employing a centrifuging piston cylinder and application to layered mafic intrusions". In: *Contributions to Mineralogy and Petrology* 168, p. 1091.

Mathez, E. A., V. J. Dietrich, J. R. Holloway, and A. E. Boudreau (1989). "Carbon Distribution in the Stillwater Complex and Evolution of Vapor During Crystallization of Stillwater and Bushveld Magmas". In: *Journal of Petrology* 30, pp. 153–173.

McBirney, A. R. (1996). "The Skaergaard Intrusion". In: *Layered Intrusions*. Ed. by R. G. Cawthorn. Vol. 15. Developments in Petrology. Amsterdam: Elsevier, pp. 147–180.

McBirney, A. R. and R. M. Noyes (1979). "Crystallization and Layering of the Skaergaard Intrusion". In: *Journal of Petrology* 20, pp. 487–554.

McCandless, T. E., J. Ruiz, B. I. Adair, and C. Freydier (1999). "Re-Os isotope and Pd/Ru variations in chromitites from the Critical Zone, Bushveld Complex, South Africa". In: *Geochimica et Cosmochimica Acta* 63, pp. 911–923.

McDonald, I., D. A. Holwell, and P. E. B. Armitage (2005). "Geochemistry and mineralogy of the Platreef and "Critical Zone" of the northern lobe of the Bushveld Complex, South Africa: implications for Bushveld stratigraphy and the development of PGE mineralisation". In: *Mineralium Deposita* 40, pp. 526–549.

Mitchell, A. A., H. V. Eales, and F. J. Krueger (1998). "Magma replenishment, and the significance of poikilitic textures, in the Lower Main Zone of the western Bushveld Complex, South Africa". In: *Mineralogical Magazine* 62, pp. 435–450.

Mitchell, A. A. and R. Manthree (2002). "The Giant Mottled Anorthosite: a transitional sequence at the top of the Upper Critical Zone of the Bushveld Complex". In: *South African Journal of Geology* 105, pp. 15–24.

Mitchell, A. A. and R. N. Scoon (2007). "The Merensky Reef at Winnaarshoek, Eastern Bushveld Complex: A Primary Magmatic Hypothesis Based on a Wide Reef Facies". In: *Economic Geology* 102, pp. 971–1009.

– (2023). "The Tennis Ball Marker in the south-eastern Bushveld Complex: comparison with the Boulder Bed of the Western Bushveld Complex and proposed genesis by disaggregation of intrusive sills in processes akin to those involved in the formation of peperites and mafic enclave swarms". In: *South African Journal of Geology* 126, pp. 407–424.

Mitchell, A. A., R. N. Scoon, and M. R. Sharpe (2019). "The Upper Critical Zone in the Swartklip Sector, north-western Bushveld Complex, on the farm Wilgerspruit 2JQ: II. Origin by intrusion of ultramafic sills with concomitant partial melting of host norite-anorthosite cumulates". In: *South African Journal of Geology* 122, pp. 143–162.

Molyneux, T. G. (2008). *A Geological Investigation of the Bushveld Complex in Sekhukhuneland and Part of the Steelpoort Valley.*

Mondal, S. K. and E. A. Mathez (2007). "Origin of the UG2 chromitite layer, Bushveld Complex". In: *Journal of Petrology* 48, pp. 495–510.

Mukherjee, R., R. Latypov, and A. Balakrishna (2017). "An intrusive origin of some UG-1 chromitite layers in the Bushveld Igneous Complex, South Africa: Insights from field relationships". In: *Ore Geology Reviews* 90, pp. 94–109.

Mungall, J. E., S. L. Kamo, and S. McQuade (2016). "U–Pb geochronology documents out-of-sequence emplacement of ultramafic layers in the Bushveld Igneous Complex of South Africa". In: *Nature Communications* 7, p. 13385.

Nomade, S., P. R. Renne, and R. K. W. Merkle (2004). "40Ar/39Ar age constraints on ore deposition and cooling of the Bushveld Complex, South Africa". In: *Journal of the Geological Society* 161, pp. 411–420.

Paterson, S. R. (2009). "Magmatic tubes, pipes, troughs, diapirs, and plumes: Late-stage convective instabilities resulting in compositional diversity and permeable networks in crystal-rich magmas of the Tuolumne batholith, Sierra Nevada, California". In: *Geosphere* 5, pp. 496–527.

Penniston-Dorland, S. C., E. A. Mathez, B. A. Wing, J. Farquhar, and J. A. Kinnaird (2012). "Multiple sulfur isotope evidence for surface-derived sulfur in the Bushveld Complex". In: *Earth and Planetary Science Letters* 337–338, pp. 236–242.

Pollard, D. D., O. H. Muller, and D. R. Dockstader (1975). "The Form and Growth of Fingered Sheet Intrusions". In: *GSA Bulletin* 86, pp. 351–363.

Richardson, S. H. and S. B. Shirey (2008). "Continental mantle signature of Bushveld magmas and coeval diamonds". In: *Nature* 453, pp. 910–913.

Roberts, M. D. et al. (2007). "The Merensky Cyclic Unit and its impact on footwall cumulates below Normal and Regional Pothole reef types in the Western Bushveld Complex". In: *Mineralium Deposita* 42, pp. 271–292.

Roelofse, F. and L. D. Ashwal (2012). "The Lower Main Zone in the Northern Limb of the Bushveld Complex–a >1.3 km Thick Sequence of Intruded and Variably Contaminated Crystal Mushes". In: *Journal of Petrology* 53, pp. 1449–1476.

Schiffries, C. M. (1982). "The petrogenesis of a platiniferous dunite pipe in the Bushveld Complex; infiltration metasomatism by a chloride solution". In: *Economic Geology* 77, pp. 1439–1453.

Schiffries, C. M. and D. M. Rye (1989). "Stable Isotopic Systematics of the Bushveld Complex; I, Constraints of Magmatic Processes in Layered Intrusions". In: *American Journal of Science* 289, pp. 841–873.

– (1990). "Stable Isotopic Systematics of the Bushveld Complex: II. Constraints on Hydrothermal Processes in Layered Intrusions". In: *American Journal of Science* 290, pp. 209–245.

Schoenberg, R., F. J. Kruger, T. F. Nagler, T. Meisel, and J. D. Kramers (1999). "PGE Enrichment in Chromitite Layers and the Merensky Reef of the Western Bushveld Complex; a Re–Os and Rb–Sr Isotope Study". In: *Earth and Planetary Science Letters* 172.1, pp. 49–64. DOI: 10.1016/S0012-821X(99)00198-3.

Scoates, J. S., C. J. Wall, R. M. Friedman, D. Weis, E. A. Mathez, and J. A. VanTongeren (2021). "Dating the Bushveld Complex: Timing of Crystallization, Duration of Magmatism, and Cooling of the World's Largest Layered Intrusion and Related Rocks". In: *Journal of Petrology* 62.2, egaa107. DOI: 10.1093/petrology/egaa107.

Scoon, R. N. and A. A. Mitchell (1994). "Discordant Iron-Rich Ultramafic Pegmatites in the Bushveld Complex and their Relationship to Iron-Rich Intercumulus and Residual Liquids". In: *Journal of Petrology* 35, pp. 881–917.

Scoon, R. N. and A. A. Mitchell (2012). "The Upper Zone of the Bushveld Complex at Roossenekal, South Africa: Geochemical Stratigraphy and Evidence of Multiple Episodes of Magma Replenishment". In: *South African Journal of Geology* 115, pp. 515–534.

– (2023). "Petrogenesis of the Marginal Sill Phase and Lower Zone in the Eastern Limb of the Bushveld Complex: Incremental development of igneous layering and syn-magmatic emplacement of peridotite intrusions and sills". In: *South African Journal of Geology* 126, pp. 425–450.

Scoon, R. N. and B. Teigler (1994). "Platinum-group element mineralization in the critical zone of the western Bushveld Complex; I, Sulfide poor-chromitites below the UG-2". In: *Economic Geology* 89, pp. 1094–1121.

Seabrook, C. L., R. G. Cawthorn, and F. J. Kruger (2005). "The Merensky Reef, Bushveld Complex: Mixing of Minerals Not Mixing of Magmas". In: *Economic Geology* 100, pp. 1191–1206.

Setera, J. B. and J. A. VanTongeren (2018). "Lateral Variability in the Upper Main Zone, Bushveld Complex, owing to Directional Magma Recharge and Emplacement from North to South". In: *Journal of Petrology* 59, pp. 1763–1786.

Sharpe, M. R. (1985). "Strontium isotope evidence for preserved density stratification in the main zone of the Bushveld Complex, South Africa". In: *Nature* 316, pp. 119–126.

Sharpe, M. R. and B. Chadwick (1982). "Structures in Transvaal Sequence Rocks within and adjacent to the Eastern Bushveld Complex". In: *Transactions of the Geological Society of South Africa* 85, pp. 29–41.

Sharpe, M. R. and L. J. Hulbert (1985). "Ultramafic sills beneath the eastern Bushveld Complex; mobilized suspensions of early lower zone cumulates in a parental magma with boninitic affinities". In: *Economic Geology* 80, pp. 849–871.

Smith, W. D. et al. (2023). "New perspectives on the formation of the Boulder Bed of the western Bushveld Complex, South Africa". In: *Mineralium Deposita* 58, pp. 617–638.

Sonnenthal, E. L. (1992). "Geochemistry of dendritic anorthosites and associated pegmatites in the Skaergaard Intrusion, East Greenland: Evidence for metasomatism by a chlorine-rich fluid". In: *Journal of Volcanology and Geothermal Research* 52, pp. 209–230.

South African Committee for Stratigraphy (1980). *Stratigraphy of South Africa. Part 1: Lithostratigraphy of the Republic of South Africa, South West Africa/Namibia, and the Republics of Bophuthatswana, Transkei and Venda*. Handbook of the Geological Survey of South Africa 8. Pretoria: Government Printer.

Stewart, B. W. and D. J. DePaolo (1990). "Isotopic studies of processes in mafic magma chambers: II. The Skaergaard Intrusion, East Greenland". In: *Contributions to Mineralogy and Petrology* 104, pp. 125–141.

van Biljon, S. (1949). "The transformation of the Pretoria Series in the Bushveld Complex". In: *Transactions of the Geological Society of South Africa* 52, pp. 1–198.

van der Merwe, M. (2007). "The occurrence of the critical zone along the exposed southeastern sector of the eastern Bushveld Complex". In: *South African Journal of Geology* 110, pp. 617–630.

van der Merwe, M. J. (1976). "The Layered Sequence of the Potgietersrus Limb of the Bushveld Complex". In: *Economic Geology* 71, pp. 1337–1351.

VanTongeren, J. A. and E. A. Mathez (2013). "Incoming Magma Composition and Style of Recharge below the Pyroxenite Marker, Eastern Bushveld Complex, South Africa". In: *Journal of Petrology* 54, pp. 1585–1605.

VanTongeren, J. A., E. A. Mathez, and P. B. Kelemen (2010). "A Felsic End to Bushveld Differentiation". In: *Journal of Petrology* 51, pp. 1891–1912.

Viljoen, M. J. and R. N. Scoon (1985). "The distribution and main geologic features of discordant bodies of iron-rich ultramafic pegmatite in the Bushveld Complex". In: *Economic Geology* 80, pp. 1109–1128.

von Gruenewaldt, G., M. R. Sharpe, and C. J. Hatton (1985). "The Bushveld Complex; introduction and review". In: *Economic Geology* 80, pp. 803–812.

Voordouw, R., J. Gutzmer, and N. J. Beukes (2009). "Intrusive origin for Upper Group (UG1, UG2) stratiform chromitite seams in the Dwars River area, Bushveld Complex, South Africa". In: *Mineralogy and Petrology* 97, pp. 75–94.

Wager, L. R., G. M. Brown, and W. J. Wadsworth (1960). "Types of Igneous Cumulates". In: *Journal of Petrology* 1, pp. 73–85.

Wall, C. J. et al. (2018). "The Stillwater Complex: Integrating Zircon Geochronological and Geochemical Constraints on the Age, Emplacement History and Crystallization of a Large, Open-System Layered Intrusion". In: *Journal of Petrology* 59, pp. 153–190.

Walraven, F. (1985). "Genetic aspects of the granophyric rocks of the Bushveld Complex". In: *Economic Geology* 80, pp. 1166–1180.

Webb, S. J., R. G. Cawthorn, T. Nguuri, and D. James (2004). "Gravity modeling of Bushveld Complex connectivity supported by Southern African Seismic Experiment results". In: *South African Journal of Geology* 107, pp. 207–218.

Willmore, C. C., A. E. Boudreau, and F. J. Kruger (2000). "The Halogen Geochemistry of the Bushveld Complex, Republic of South Africa: Implications for Chalcophile Element Distribution in the Lower and Critical Zones". In: *Journal of Petrology* 41, pp. 1517–1539.

Willmore, C. C., A. E. Boudreau, A. Spivack, and F. J. Kruger (2002). "Halogens of Bushveld Complex, South Africa: δ^{37}Cl and Cl/F evidence for hydration melting of the source region in a back-arc setting". In: *Chemical Geology* 182, pp. 503–511.

Wilson, A. H. (2012). "A Chill Sequence to the Bushveld Complex: Insight into the First Stage of Emplacement and Implications for the Parental Magmas". In: *Journal of Petrology* 53, pp. 1123–1168.

Wilson, A. H., A. Zeh, and A. Gerdes (2017). "In Situ Sr isotopes in Plagioclase and Trace Element Systematics in the Lowest Part of the Eastern Bushveld Complex: Dynamic Processes in an Evolving Magma Chamber". In: *Journal of Petrology* 58, pp. 327–360.

Wilson, H. D. B. (1956). "Structure of Lopoliths". In: *GSA Bulletin* 67, pp. 289–300.

Worst, B. G. (1958). "The Differentiation and Structure of the Great Dyke of Southern Rhodesia". In: *South African Journal of Geology* 61.1, pp. 283–358.

Wotzlaw, J.-F., U. Schaltegger, D. A. Frick, M. A. Dungan, A. Gerdes, and D. Günther (2013). "Tracking the Evolution of Large-Volume Silicic Magma Reservoirs from Assembly to Supereruption". In: *Geology* 41, pp. 867–870.

Yao, Z., J. E. Mungall, and M. C. Jenkins (2021). "The Rustenburg Layered Suite formed as a stack of mush with transient magma chambers". In: *Nature Communications* 12, p. 505.

Yuan, Q. et al. (2017). "Pulses of Plagioclase-laden Magmas and Stratigraphic Evolution in the Upper Zone of the Bushveld Complex, South Africa". In: *Journal of Petrology* 58, pp. 1619–1643.

Yudovskaya, M. A. et al. (2021). "Hybrid norite and the fate of argillaceous to anhydritic shales assimilated by Bushveld melts". In: *Mineralium Deposita* 56, pp. 73–90.

Zeh, A., M. Ovtcharova, A. H. Wilson, and U. Schaltegger (2015). "The Bushveld Complex was emplaced and cooled in less than one million years – results of zirconology, and geotectonic implications". In: *Earth and Planetary Science Letters* 418, pp. 103–114.

Zirakparvar, N. A., E. A. Mathez, J. S. Scoates, and C. J. Wall (2014). "Zircon Hf Isotope Evidence for an Enriched Mantle Source for the Bushveld Igneous Complex". In: *Contributions to Mineralogy and Petrology* 168, p. 1050.